

Tax Reform, State Capacity, and Inherited Distortions: Officer Discretion and Property-Tax Progressivity in Pakistan

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Abstract

I study how assessment incentives shape property records and how those records persist across valuation reform in Punjab, Pakistan. A model of officers' assessment choices shows why constraining departures from valuation rules need not correct inherited input errors. I combine a difference-in-discontinuities around a historical exemption threshold, five years of administrative records, and liability comparisons holding the tax formula fixed. The discontinuity in recording at or below five marla is 66.6 percentage points larger where the exemption applied. Among properties with valid land records throughout the panel, 92.3% retain unchanged areas. Replacing administrative measurements with satellite measurements under the same formula raises the Kakwani index by 0.0504. Discrepancy-based targeting also identifies more field-confirmed errors than random inspection. These findings show how inherited information can transmit earlier assessment distortions into a new tax system, making verification and record correction complements to valuation reform.

1 Introduction

Sustained development requires governments to raise revenue to fund public services and infrastructure. Financing local investments such as water and sanitation through property taxes offers clear benefits. Recurrent immovable property taxes are relatively efficient because their tax base is hard to move, making them less distortionary than many other taxes (Norregaard, 2013; Mirrlees et al., 2011, p.368-405).

Yet, property taxation is heavily underused in developing economies. In Punjab, Pakistan, property tax-to-GDP ratios are only 0.04%–0.07%, far below the 0.30%–0.42% average in comparable low- and middle-income countries (Bank, 2022). This weak revenue base strains fast-growing cities such as Lahore and Multan, where about 35%–40% of urban households lack basic services including piped water, waste management, and functional sanitation. Property tax is therefore not just an administrative levy, but a key tool to close widening municipal financing gaps. Similarly, real estate has long been a major channel for elite wealth accumulation, yet the province captures less than 0.07% of the total capital value of registered immovable assets (Centre, 2023). Shifting from regressive consumption taxes toward a more progressive property tax can help capture unearned land value gains, requiring wealthier asset owners to contribute more and helping finance public goods for underserved urban residents.

To address these chronic inefficiencies, recent province-level reforms have explicitly targeted the structural valuation system. Whether these changes deliver their intended gains, however, depends on how far reform overcomes institutional persistence in the assessment process. This is because property attributes used for valuation can persist across successive systems without being remeasured. If earlier tax incentives encouraged underassessment, the resulting records may carry those distortions into a new system even when its valuation rules are correctly applied. How do tax incentives shape the recorded property-tax base, and how does the persistence of those records affect the implementation of a new valuation system?

I study this question through a large-scale property-tax reform in Punjab, Pakistan, which shifted the valuation basis from annual rental value to capital value. The reform aimed to make revenue more responsive to rising property values and help finance growing demand for public services (Sub-National Governance Programme, 2025). The earlier system calculated liabilities using property characteristics and standardised valuation rates (Khan et al., 2016a). The new system applies published valuation rates to recorded land and covered area while retaining existing records of these attributes. This continuity allows me to examine how information produced under earlier assessment incentives shapes the implementation of new valuation rules. I ask three linked questions. First, how do tax incentives shape the property attributes entered into administrative records? Second, to what extent do these records persist across the change in valuation rules? Third, how do inherited measurements affect liabilities under the new system, and can independent information help identify records that require correction?

I answer these questions by developing a model of officers' assessment choices under competing incentives from informal payments, revenue collection and administrative scrutiny. The model distinguishes between departures from the prescribed valuation formula and errors already embedded in the property records used by that formula. I then examine this mechanism using three empirical methods. First, I estimate a difference-in-discontinuities around the historical five-marla exemption threshold, comparing localities where the exemption applied with those where it did not, using satellite-measured area to locate properties around the threshold. This tests whether exemption incentives affected administrative recording. Second, I use a five-year property panel to measure the persistence of recorded attributes across the valuation reform. Third, I calculate counterfactual liabilities using administrative and satellite measure-

ments under the same statutory formula and compare their progressivity relative to a common property-value distribution. I complement these analyses with field validation to assess measurement accuracy and the targeting of verification, and interviews with 300 frontline officials to describe how property information is produced and entered into administrative records.

I find that the jump in the probability of being recorded at or below five marla is 66.6 percentage points larger in localities where the exemption applied than in those where it did not (standard error: 16.6 percentage points). Among properties with valid land-area records in all five years, 92.3% retain exactly the same recorded area, so the new valuation system largely inherits the earlier physical records. Applying the same tax formula to satellite rather than administrative measurements raises the Kakwani progressivity index by 0.0504 (95% confidence interval: 0.0141–0.0992), holding the predicted property-value distribution fixed. Finally, among the 10% of properties with the largest administrative–satellite land-area discrepancies, 77.9% have a field-confirmed error, compared with 39.5% under random selection. These findings imply that tax reform can inherit the consequences of earlier assessment incentives through the records used to implement new rules. Making valuation calculations more transparent can therefore leave distortions in the tax base, limiting the progressivity achieved under the new schedule. Independent measurement can help direct scarce verification resources towards inaccurate records, but realising its benefits requires an administrative process that translates verified discrepancies into corrected assessments.

The paper contributes to the literature on institutional persistence by examining the information through which formal rules operate. North (1993) emphasises that changes in formal rules need not transform the broader constraints governing implementation. Acemoglu and Robinson (2008) show how persistence in effective power can offset formal institutional change. I examine a different channel: the continued use of administrative property characteristics across valuation regimes. The panel documents continuity, while the common-formula comparison shows how the choice of physical inputs shapes liabilities under the new rules. This connects institutional persistence to a measurable feature of everyday administration without attributing record stability to a particular political mechanism.

The closest tax-administration studies examine incentives, reporting responses and assessment disparities. Khan et al. (2016b) study performance incentives for Punjab tax collectors; Best et al. (2015) and Carrillo et al. (2017) examine how tax design and enforcement interact with reporting behaviour. Avenancio-León and Howard (2022) show how property assessment generates unequal effective burdens. This paper focuses on the continuity of physical inputs across a change in valuation rules. Its contribution is to connect evidence on recording and persistence with the distribution of liabilities calculated from those inputs.

The technology evidence concerns an independent check on this inherited information. Knebelmann et al. (2026) experimentally compare bureaucratic and algorithmic property valuation in Senegal. Aman-Rana and Minaudier (2026) show that digitising land records in Punjab can weaken tax collection through changes in bureaucratic capacity. These studies make clear that the consequences of technology depend on what it changes in administration. Here, remote sensing supplies measurements against which existing records can be checked. The targeting exercise evaluates that information’s usefulness for verification, while correction remains the next administrative step.

Section 2 explains the change in valuation rules and the records it inherited. Section 3 provides the conceptual framework. Sections 4 and 5 describe the data and empirical comparisons. Section 6 presents measurement validation, recording incentives, persistence and distributional consequences, together with the verification application. Section 7 examines sensitivity and scope, and Sections 8 and 9 draw the implications for institutional reform.

2 Institutional Setting

2.1 Property-value Assessment and the Role of Front-line Tax Officers

Punjab’s Urban Immovable Property Tax (UIPT) is administered by the provincial Excise and Taxation Department. The department divides the urban jurisdictions subject to UIPT into a five-level geographical hierarchy, illustrated in Figure A.1. The primary unit for property inspection, assessment, collection, and enforcement is called the tax “circle”. Each circle is assigned an inspector, supported by a clerk and a constable. The inspector leads property visits, determines the relevant property characteristics, and takes primary responsibility for the recommended assessment. The clerk and constable assist with measurement, documentation, record preparation, and the delivery of notices. Accordingly, the term “officer” in the theoretical analysis refers to the inspector as the relevant front-line decision-maker.

The circle team cannot enter its recommendation directly into the official tax record. It must submit the assessment to the Excise and Taxation Officer (ETO), whose approval is mandatory. The ETO reviews the measurements, classifications, and supporting record prepared by the inspector-led team. The process therefore separates informational control from formal authority: the inspector controls the production of property-specific information, while the ETO controls whether the recommendation receives formal approval. For analytical tractability, the model assumes that the ETO is not a party to any informal bargain between the inspector and the taxpayer. ETO approval nevertheless matters because the inspector must produce a record that is sufficiently complete and internally consistent to obtain formal approval.

2.2 The Capital-value Reform

Before 2025, UIPT liability was based on annual rental value (ARV). The inspector recommended a rental value by applying the department’s notified valuation tables, reproduced in Figure A.2, to the property’s location, use, occupancy, condition, and physical characteristics. Although the resulting rental value appeared in the administrative record and informed the taxpayer’s bill, the taxpayer could not independently determine the correct rental value from publicly observable information. The tables constrained assessments but left substantial room for judgement about prevailing rents and property characteristics (Piracha and Moore, 2015; Khan et al., 2016b). Consequently, a low assessment could reflect either a legitimate valuation judgement or an officer-mediated departure, and the administrative record alone could not distinguish between them.

Figure A.3 shows the Punjab Finance Act 2024 Amendment, which replaced annual rental value (ARV) with capital value (CV) as the statutory valuation basis. The change took effect on 1 January 2025, during fiscal year 2024–25. Under CV, recorded land and covered area are valued using the applicable district-collector neighbourhood rates, referred to below as DC rates. The resulting capital value is taxed under the value-band schedule illustrated in Figures A.4.

2.3 Verifiability and the Inherited Cadastre

The reform changed both the valuation basis and the information required to support an assessment. Under ARV, the recorded value depended on judgements that a taxpayer or supervising officer could not reproduce independently. Under CV, the statutory calculation combines recorded physical attributes with published rates. The relevant property characteristics and CV details appear in the administrative record and on the converted taxpayer’s bill. A taxpayer,

supervisor, or other third party can therefore reproduce the official calculation and determine whether it is internally consistent with the recorded inputs. Figure A.5 illustrates this change in the information displayed on the bill.

The reform did not, however, introduce a general remeasurement of properties. The land area and covered area used under CV were inherited from the ARV-era cadastre. It also left the audit budget, inspection resources, officer remuneration and benefits, sanction schedule, frequency of review, and disciplinary chain substantially unchanged. Qualitative evidence from officials confirms that the principal institutional change concerned the valuation rule rather than enforcement intensity or officer incentives outside the assessment process.

The distinction between verifiability and accuracy is therefore central. A CV assessment can be reproduced exactly from the official record even when the recorded land or covered area is stale or incorrect. Greater verifiability may induce officers to correct some records because inconsistencies are easier for taxpayers and supervisors to detect, but the formula does not mechanically correct its own inputs. Field measurements are then used to determine whether remote sensing provides a more accurate benchmark than the administrative record.

3 Theoretical Framework

3.1 The Officer's Payoff and Optimal Departure

Qualitative surveys conducted with frontline tax officers and literature on frontline tax officers suggest that the officer trades a public benefit of raising revenue for the state against a private benefit of taking a bribe for himself.

The officer's choice. For each property i under valuation regime $r \in \{\text{ARV}, \text{CV}\}$, let $\bar{A}_{ir}$ denote the assessment that would be recorded if the officer assesses the property correctly. This is the official, regime-specific benchmark relevant to the officer's decision. It is determined under the ARV procedure before the reform and under the statutory CV rule after the reform.

The officer chooses to under-assess denoted by $d_{ir} \geq 0$. The assessment resulting from this choice is

$$A_{ir} = \bar{A}_{ir} - d_{ir}. \quad (3.1)$$

Thus, $d_{ir} = 0$ means that the value is recorded correctly, while $d_{ir} > 0$ means that the recorded assessment is under-assessed. The object d_{ir} is the strategic choice analysed in the officer's payoff problem.

Assumption 1. Throughout this paper, I assume that the officer chooses to underassess from the true value such that $d_{ir} \geq 0$ and $A_{ir} \leq \bar{A}_{ir}$.

Let $T_r(\cdot)$ denote statutory tax liability. The officer's payoff is

$$\Pi_{ir}(d_{ir}) = \mu T_r(\bar{A}_{ir} - d_{ir}) + \theta [T_r(\bar{A}_{ir}) - T_r(\bar{A}_{ir} - d_{ir})] - \gamma_r d_{ir}^2 - \delta p_{ir}(d_{ir}). \quad (3.2)$$

Here, $\mu \in [0, 1)$ is the inspector's weight on recorded revenue—the margin affected by the Punjab performance-incentive experiments (Khan et al., 2016b, 2019). The parameter θ is the inspector's marginal keep-share, $\delta \geq 0$ is the inspector's loss conditional on an audit, and $p_{ir}(\cdot)$ is the probability that a departure attracts scrutiny.

Maintained assumptions. The tax schedule is finite, nonnegative and nondecreasing, and the officer's maximum is attained. Scrutiny occurs with probability $p_{ir}(d_{ir}) = \bar{p}(d_{ir}/\bar{A}_{ir})^2$, where

$\bar{p} \in [0, 1]$. The verifiability comparison holds the record, benchmark, tax schedule, θ , μ , δ , and $\bar{p}$ fixed. The FOC below applies to a positive interior optimum within an affine tax segment of marginal rate $\tau > 0$; it does not impose a globally proportional tax schedule.

Define $H_{ir} = \gamma_r + \delta \bar{p} / (\bar{A}_{ir})^2 > 0$. Within that segment, the FOC and its solution are

$$\frac{\partial \Pi_{ir}}{\partial d_{ir}} = (\theta - \mu)\tau - 2H_{ir}d_{ir} = 0 \implies d_{ir}^* = \frac{(\theta - \mu)\tau}{2H_{ir}}. \quad (3.3)$$

The second derivative is $-2H_{ir} < 0$. A positive interior solution requires $\theta > \mu$; if $\theta \leq \mu$, the optimum is zero. Appendix B.1 derives the zero-departure corner, proves monotonicity and convergence, and treats bounds and tax-band crossings.

CV bills disclose the district-collector (DC) rates and recorded land and covered area used in the calculation. I represent this greater ability to check calculations by $\gamma_{CV} > \gamma_{ARV}$, with assessment units normalized consistently. On the interior branch,

$$\frac{\partial d_{ir}^*}{\partial \gamma_r} = -\frac{(\theta - \mu)\tau}{2H_{ir}^2} < 0. \quad (3.4)$$

3.2 Verifiability and Inherited Records

Let $X_i^\dagger = (a_i^\dagger, s_i^\dagger)$ denote correct plot and covered areas. With the CV formula h and common locality schedule S_l , define

$$V_i^\dagger = h(S_l, X_i^\dagger), \quad V_{it}^{\text{admin}} = h(S_l, X_{it}^{\text{admin}}) = V_i^\dagger + \xi_{it}^X. \quad (3.5)$$

The signed input error ξ_{it}^X can be positive or negative. These are statutory formula values, not necessarily market values. Under CV, $\bar{A}_{it,CV} = V_{it}^{\text{admin}}$, so

$$A_{it,CV}(\gamma_{CV}) = V_i^\dagger + \xi_{it}^X - d_{it,CV}^*(\gamma_{CV}). \quad (3.6)$$

Proposition 1 (Inherited error survives verifiability). *Holding the inherited record and other primitives fixed,*

$$\gamma_{CV} \rightarrow \infty \implies d_{it,CV}^* \rightarrow 0 \implies A_{it,CV} \rightarrow V_{it}^{\text{admin}} = V_i^\dagger + \xi_{it}^X. \quad (3.7)$$

The assessment converges to the value implied by the inherited record. That record need not be accurate. The result also applies directly if calculations are already mechanical and $d^* = 0$. Verifiability eliminates the officer's departure from the administrative formula value without correcting the attributes from which that value is constructed.

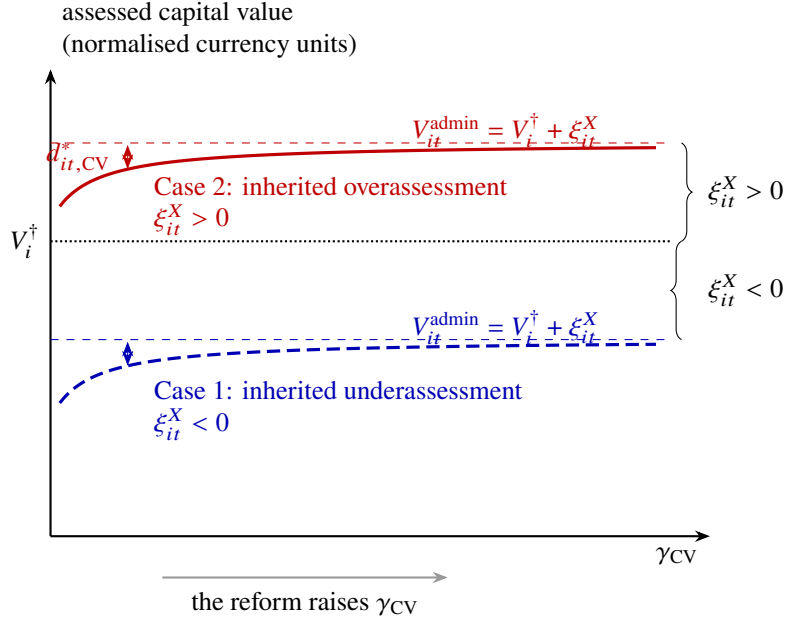


Figure 3.1: The dragged base (Proposition 1). The single panel shows two separate properties. In Case 1, inherited cadastral error is negative, so the recorded assessment remains below the correct formula value $V_i^\dagger$ and converges to an underassessed administrative value. In Case 2, inherited error is positive, so the plotted assessment lies above $V_i^\dagger$ and converges to an overassessed administrative value. In both cases $d_{it,CV}^*$ vanishes as verifiability rises, while ξ_{it}^X survives. A positive inherited error could at some finite γ_{CV} be exactly offset by $d_{it,CV}^*$; such equality would be accidental cancellation, not accurate measurement. All vertical quantities are capital values in the same currency units; the plotted numerical scale is normalised and illustrative. Verifiability is not accuracy.

3.3 Implications for Progressivity

Suppress time and CV subscripts, write $V_i = V_i^\dagger > 0$, and hold the liability schedule T fixed. The effective tax rate e_i is expressed as the total liability $T(V_i)$ as a share of property value V_i . Expressing the effective tax rate for administrative records with measurement error ξ_i^X and officer's under assessment d_i^*

$$e_i(\gamma)^{admin} = \frac{T(V_i + \xi_i^X - d_i^*(\gamma))}{V_i}, \quad e_i^{accurate} = \frac{T(V_i)}{V_i}. \quad (3.8)$$

For an exact additive decomposition and the same within-band restriction as the officer model: the correct value V_i , inherited benchmark $V_i + \xi_i^X$, and assessments under comparison all lie in a single affine segment of T , with marginal rate $\tau_i \geq 0$. It follows that

$$e_i(\gamma)^{admin} = e_i^{accurate} + \underbrace{\tau_i \frac{\xi_i^X}{V_i}}_{\text{inherited error}} - \underbrace{\tau_i \frac{d_i^*(\gamma)}{V_i}}_{\text{current discretion}}. \quad (3.9)$$

Applying Proposition 1 to Equation (3.9) gives

$$e_i^\infty \equiv \lim_{\gamma \rightarrow \infty} e_i(\gamma) = e_i^{\text{accurate}} + \tau_i \frac{\xi_i^X}{V_i}. \quad (3.10)$$

Verifiability removes the current-discretion component of the effective tax-rate distortion. It leaves the inherited-error component.

Consider representative properties $b \in \{L, M, H\}$, with $V_L < V_M < V_H$. Each may face a different statutory marginal rate, provided its own correct value, inherited benchmark and assessments under comparison remain within the same affine segment. For positive liabilities, define relative progression between two bands by

$$P_{jk}(\gamma) = \frac{e_j(\gamma)}{e_k(\gamma)}, \quad j > k. \quad (3.11)$$

This comparison is progressive when $P_{jk} > 1$, proportional when $P_{jk} = 1$, and regressive when $P_{jk} < 1$. A larger ratio means greater relative progression between the same two property values.

To derive the comparison that drives Proposition 2, divide Equation (3.10) by $e_i^{\text{accurate}} = T(V_i)/V_i > 0$:

$$\frac{e_i^\infty}{e_i^{\text{accurate}}} = 1 + \frac{\tau_i \xi_i^X / V_i}{T(V_i) / V_i} = 1 + \frac{\tau_i \xi_i^X}{T(V_i)}. \quad (3.12)$$

The property-value denominator cancels. The remaining fraction is the liability change caused by inherited error, as a share of correct-input liability. Taking the normalized rate for property j and dividing by that for property k gives the final expression below.

Proposition 2 (Inherited error and relative progressivity). *Under the within-segment restriction, with positive correct-input and limiting inherited liabilities,*

$$\frac{P_{jk}^\infty}{P_{jk}^{\text{accurate}}} = \frac{e_j^\infty / e_j^{\text{accurate}}}{e_k^\infty / e_k^{\text{accurate}}} = \frac{1 + \tau_j \xi_j^X / T(V_j)}{1 + \tau_k \xi_k^X / T(V_k)}. \quad (3.13)$$

Inherited records lower, preserve, or raise relative progressivity according as $\tau_j \xi_j^X / T(V_j)$ is smaller than, equal to, or larger than $\tau_k \xi_k^X / T(V_k)$.

Within proportional segments, the comparison reduces to the ordering of ξ_j^X / V_j and ξ_k^X / V_k . A larger proportional shortfall at the higher-value property lowers relative progressivity. The model also permits the opposite result. Less progressive does not necessarily mean regressive, and greater progressivity does not establish greater accuracy.

Appendix B.2 provides the proof and the full-schedule comparison when measurements cross tax boundaries. Appendix B.3 derives group effective rates and the Kakwani comparison. A pairwise result alone does not determine the aggregate sign. These comparisons hold inputs fixed as verifiability changes.

From the framework to the empirical exercises. The framework distinguishes applying a valuation rule from verifying the information it uses. Proposition 1 shows why closer compliance can leave inherited input discrepancies intact. Proposition 2 shows that their distributional consequences depend on how those discrepancies vary across properties. I examine three empirical aspects of this channel: whether earlier tax incentives are reflected in recording, whether

recorded attributes persist, and how alternative physical measurements change liabilities under a common formula. The first exercise concerns the possible origins of recorded discrepancies, the second establishes continuity, and the third measures their distributional relevance under the new rules. I first assess the independent measurements against field observations and also test whether they can direct verification towards records with field-confirmed errors.

4 Data

The empirical analysis combines an administrative property panel, independently produced remote-sensing measurements, field measurements, and interviews with tax officials. The administrative panel records the attributes and assessed values observed by the tax authority. Remote sensing provides an independent measurement of land and covered area, while the field data provide an external benchmark for evaluating both sources. The three property-level datasets are linked using the official property identification number (PIN).

4.1 Administrative Property Panel

The principal administrative dataset is a representative sample of 20,000 properties drawn from the official cadastre maintained by the Punjab Excise and Taxation Department (E&T) for Lahore, Punjab.

The underlying cadastre contains approximately one million properties from Lahore. I observe the sampled properties in four annual waves, covering FY2022–23 through FY2025–26. Each property has a unique Property Identification Number (PIN), which provides an exact one-to-one linkage across waves. All 20,000 sampled properties are successfully matched in every wave, producing a balanced property-level panel.

Frontline tax officers update the cadastre before each annual billing cycle to record corrections to existing properties and the addition of newly assessed properties.

For each property-year, the cadastre reports the tax circle and locality, occupancy status (rented or owner-occupied), use classification (residential or commercial), land area, covered area, number of storeys, gross annual rental value, applicable District Collector (DC) rates, and the formula-implied DC property value. For FY2024–25 and FY2025–26, the cadastre also contains an indicator identifying whether the property had converted to capital-value taxation.

This data will be used to construct V_{it}^{admin} for each year t required for G_{it} described in Section 5.4.

4.2 Independent Remote-Sensing Data

From the administrative property data of 20,000 properties I sampled 10,132 properties for creating an independent remote-sensing data. The remote-sensing data uses ESRI Satellite imagery layer from May 2026 for Lahore. For each selected property, the research team independently drew the parcel boundary and measured the principal physical inputs into capital-value assessment. Land area is calculated from the resulting parcel polygon, while covered area is predicted from the imagery using a machine-learning model. I denote these independently measured attributes by X_i^{RS} . Details of the imagery, parcel delineation, and prediction procedure are reported in the appendix.

The remote-sensing observations link exactly to the administrative panel by PIN. Consequently, differences between X_i^{RS} and X_{it}^{admin} compare alternative measurements of the same

property rather than potentially mismatched records. Because the remote-sensing data are observed only in 2026, they provide a cross-sectional benchmark rather than an annual measure of changes in the physical property.

Sampling frame and coverage. Figure A.6 shows the layer covers 10,132 properties across 48 of Lahore’s E&T circles, drawn as a representative sample of the city’s taxable base and excluding high density (greater than 150 persons per 100 *m*), contiguous, and irregular building fabric which makes imagery-based delineation of individual structures infeasible. The population-density distribution that underlies the density comparison and the 150-person exclusion is shown in Figure E.9 in Appendix A. This is the standard limitation the remote-sensing literature documents for dense informal morphologies (Kuffer et al., 2016), and one that binds with equal force in the historic core. For each property the layer contains: the building-footprint, building height, polygon digitised from high-resolution imagery, a predicted covered area.

Coverage is spatially broad but selective on property type, and the selection is deliberate. Footprint delineation succeeds in planned, organised, and gated layouts of regularly spaced structures and fails in the unplanned fabric, so the frame was drawn from the organised segment of the city by design. Against the cadastre as a whole, the remotely-sensed sample skews toward larger plots, larger covered area, and higher liability, with normalised differences of 0.20 to 0.32 (Table 4.1), and it is 77 % organised against 30 % in the universe. Because the sample size makes conventional balance tests uninformative, with every difference being significant at $p < 0.001$, I follow Imbens and Wooldridge (2009) and provide the normalized difference, which does not grow mechanically with the sample.

I define an organized-or-gated indicator equal to one for any property recorded as organized, planned, or gated, and compare the RS sample to the organized-or-gated properties it did *not* sample. Within that segment the imbalance largely disappears: the normalized differences fall to 0.08–0.14 (Table 4.1, final column), well inside the 0.25 rule of thumb for substantial imbalance (Imbens and Wooldridge, 2009; Imbens and Rubin, 2015). Conditional on the organized city that the technology can see, the RS sample is representative; the raw imbalance against the full register is the organized-segment restriction.

Table 4.1: Balance: RS sample against the organized/gated segment and the non-selected register

	RS selected	Organized/ gated	Non- selected	Normalized difference	
				vs org./gated	vs non-selected
Land area (sq. yd)	392.7	276.3	179.4	0.14	0.25
Covered area (sq. ft)	3,497	2,723	1,736	0.09	0.20
Tax liability (Rs)	9,670	7,986	4,184	0.09	0.32
Population density (/100 m)	118.6	117.3	114.8	0.08	0.21
Observations	10,132	156,983	546,108		

Group means. “Organized/gated” is the set of *non-selected* properties recorded as organized, planned, or gated, the segment the RS frame was drawn to cover; “non-selected” is the entire register outside the RS sample. Balance is read from the normalized difference, $(\bar{x}_{RS} - \bar{x}_g) / \sqrt{(s_{RS}^2 + s_g^2)/2}$ (Imbens and Rubin, 2015), with $|n.d.| > 0.25$ signalling substantial imbalance (Imbens and Wooldridge, 2009; Imbens and Rubin, 2015). At this sample size every pairwise *t*-test rejects at $p < 0.001$, so no *p*-values are reported. Density is WorldPop 2020 per 100 m cell.

4.3 Field Measurement and Validation Sample

For 2,100 properties within the remote-sensing sample, trained enumerators independently measured land and covered area through physical site visits. I denote these field measurements by X_i^{site} . The observations are linked to the administrative and remote-sensing records by PIN, producing a sample in which the same physical attributes are observed from three sources. The field protocol, sample selection, and quality-control procedures are described in the appendix.

4.4 Interviews with Frontline Tax Officials

I also draw on interviews conducted in 2026 with 300 distinct frontline E&T officials. The interviews document the assessment and approval process, the location of discretion within the administrative hierarchy, and whether enforcement practices changed alongside the reform. They provide institutional evidence used to interpret the empirical estimates but do not determine treatment status or enter the construction of the quantitative outcomes.

5 Outcome Measures and Empirical Strategy

To answer how inherited records shape liabilities under new rules, I examine recording incentives, continuity of administrative attributes and the distributional consequences of using alternative physical inputs. These exercises play different roles: the threshold comparison addresses incentive-related recording, the panel measures persistence, and the common-formula comparison isolates the distributional contribution of measured inputs. I first evaluate remote-sensing (RS) measurements against field observations. The verification-targeting exercise then asks how this independent information can help identify records requiring attention.

5.1 Relative Measurement Performance

A third-party technology can serve as an independent benchmark only if its measurement error is smaller than that of the record it evaluates. Let $\mathcal{V}$ denote the population represented by the site-validation sample. For attribute $k \in \{\text{plot area, built-up area}\}$ and source $m \in \{\text{admin, RS}\}$, define mean-squared error against the site protocol

$$\text{MSE}_k^m = \mathbb{E}_{\mathcal{V}}[(X_{i,k}^m - X_{i,k}^{\text{site}})^2], \quad (5.1)$$

and

$$\lambda_k = 1 - \frac{\text{MSE}_k^{\text{RS}}}{\text{MSE}_k^{\text{admin}}}, \quad \lambda_k \leq 1. \quad (5.2)$$

$\lambda_k > 0$ means RS has lower mean-squared error than the administrative record; $\lambda_k = 0$ equal; $\lambda_k < 0$ worse.

Estimation and interpretation. Let n be the number of properties in the validation sample. For each source m the sample mean-squared error is

$$\widehat{\text{MSE}}_k^m = \frac{1}{n} \sum_{i=1}^n (X_{i,k}^m - X_{i,k}^{\text{site}})^2, \quad m \in \{\text{admin, RS}\}, \quad (5.3)$$

computed over the same n properties for both sources, so the two errors are measured against an identical benchmark. Then

$$\hat{\lambda}_k = 1 - \frac{\widehat{\text{MSE}}_k^{\text{RS}}}{\widehat{\text{MSE}}_k^{\text{admin}}}. \quad (5.4)$$

percentile confidence intervals come from 1,000 block-bootstrap draws that resample assessment circles with replacement and recompute (5.3)–(5.4) on each draw, preserving dependence among properties within a circle. If RS provides a useful independent benchmark, its measurement error is smaller than administrative error and $\lambda_k > 0$

5.2 Targeting Verification

I ask whether checking properties with large administrative–RS discrepancies finds more field-confirmed errors than checking properties at random. This comparison uses the validation sample, restricted to positive measurements from all three sources for the attribute being examined.

For each property, the ranking score and the error indicator are

$$Q_{i,k} = \left| \log X_{i,k}^{\text{RS}} - \log X_{i,k}^{\text{admin}} \right|, \quad E_{i,k}(t) = \mathbf{1} \left\{ \left| \log X_{i,k}^{\text{admin}} - \log X_{i,k}^{\text{site}} \right| > t \right\}. \quad (5.5)$$

The score uses administrative and RS measurements only. Field measurement determines whether the record counts as an error. The main tolerance is $t = 0.25$: the administrative measurement must be more than 28.4% above, or 22.1% below, the field measurement.

Estimation and interpretation. I rank properties by $Q_{i,k}$ and select the highest 10%. The error rate in this list is its precision. The error rate in the full eligible validation sample is the expected precision of random selection. Their difference is the targeting gain: the additional errors found per inspection. Appendix C.4 defines these quantities and the additional measures of recall and error capture.

The ranking rule is fixed without using field outcomes. I use 1,000 assessment-circle bootstrap draws, rebuilding the ranking in each draw. The comparison measures the ability to find discrepancies confirmed by the field protocol.

5.3 Recording and Tax Incentives

5.3.1 The Land-area Exemption Threshold

Under the earlier annual-rental-value (ARV) system, residential properties recorded at five marla or less qualified for a size exemption outside Category-A localities. A marla is a local land-area unit; the threshold is five such units. Category-A properties did not receive this exemption. The comparison asks whether recording changes differently at this threshold where exemption is available.

Let A_i be recorded land area, X_i RS land area, $x_i = X_i - 5$, $S_i = \mathbf{1}\{x_i > 0\}$, and $N_i = \mathbf{1}\{\text{locality is non-A}\}$. The outcome is

$$Y_i^5 = \mathbf{1}\{A_i \leq 5\}. \quad (5.6)$$

This indicates placement at or below the threshold. RS determines a property's position around the threshold but does not enter the outcome. The reported estimates use the FY2026–27 record.

Identification. I compare the jump in Y_i^5 as RS area crosses five marla in non-A localities with the corresponding jump in Category A. Their difference, d_5 , isolates exemption-related placement if the two groups would otherwise have the same discontinuity. This requires no differential change in benchmark error, property composition or sample selection at the cutoff. It also requires that actual plot sizes do not sort differently around the cutoff in response to the exemption. Category A is not randomly assigned, so these assumptions need substantive support.

Estimation and interpretation. I fit a separate straight line on each side of the threshold in each locality group:

$$Y_i^5 = \alpha + d_0 N_i + E S_i + d_5 (N_i S_i) + \beta_0 x_i + \beta_1 (N_i x_i) + \beta_2 (S_i x_i) + \beta_3 (N_i S_i x_i) + \Gamma' Z_i + \varepsilon_i. \quad (5.7)$$

Because Y_i^5 is binary, this is a local linear probability regression; level coefficients are %-point differences. The intercept α is the predicted exemption rate for a Category-A property at the five-marla benchmark approached from below. The indicator N_i equals one for a non-A locality, so d_0 is the non-A–Category-A difference at that same left limit. The indicator S_i equals one above the five-benchmark marla, so $E = E_A$ is the Category-A discontinuity. The product $N_i S_i$ equals one only for non-A properties above the cutoff; its coefficient d_5 is the additional non-A discontinuity which is the coefficient of interest. The non-A discontinuity is $E + d_5$, and the causal difference in discontinuities is $(E + d_5) - E = d_5$, under the assumptions below. A positive value means that crossing the benchmark threshold produces a more favourable change in recorded placement where the exemption applies. Because the outcome is binary, multiplying d_5 by 100 expresses it in %-points.

The running variable $x_i = X_i - 5$ measures RS land area in marla relative to the cutoff. The coefficient β_0 is the Category-A slope below five; β_1 is the additional non-A slope below five; β_2 is the change in the Category-A slope above five; and β_3 is the additional change in the non-A slope above five. Thus all four locality-by-side cells have their own local linear trend. The vector Z_i contains predetermined precision controls for main-road location and log RS covered area, Γ is the corresponding coefficient vector, and ε_i is the remaining property-level disturbance.

I estimate Equation (5.7) over the common window $|x_i| \leq 2$, using triangular weights $w_i = (1 - |x_i|/2)\mathbf{1}\{|x_i| \leq 2\}$. The common two-marla bandwidth ensures that the Category-A and non-A discontinuities are compared over the same benchmark neighbourhood; narrower windows contain too little Category-A support to represent the behaviour of interest reliably. Inference uses 500 joint bootstrap replications that resample assessment circles and apply the same circle multiplicity to both locality-group equations, thereby preserving their covariance.

As a descriptive check, I also compare the share recorded in the favourable interval $[3, 5]$ among properties that RS places above five. This check illustrates the placement pattern but has no below-threshold comparison. Its construction is in Appendix C.6.

5.3.2 Land Understatement and Potential Tax Liability

I examine whether land understatement is more common among properties with larger potential tax liabilities. I calculate potential liability, K_i , by applying the pre-reform annual rental value (ARV) formula to site-measured land and covered areas, using recorded locality and occupancy classifications. This calculation excludes individual allowances, exemptions and rebates and does not measure tax payments. Appendix C.7 gives the formula.

Let D_i^A equal one when recorded land area is below site-measured area. I estimate

$$D_i^A = \alpha_{c(i)} + f(\log K_i) + u_i, \quad (5.8)$$

where assessment-circle fixed effects, $\alpha_{c(i)}$, make the comparisons within circles, and f is a natural spline whose flexibility is selected by the Bayesian information criterion. I plot the estimated probability of understatement against potential liability, holding assessment-circle composition fixed. Pointwise confidence intervals come from resampling assessment circles. This conditional association does not identify the effect of liability on recording.

5.4 Persistence of Recorded Attributes

I measure persistence in two ways: whether a property's recorded area stays exactly the same, and how its difference from a fixed RS benchmark changes over time. For attribute k , define

$$G_{it,k} = \log X_{i,k}^{\text{RS}} - \log X_{it,k}^{\text{admin}}. \quad (5.9)$$

Positive values mean that RS area exceeds recorded area. Because the same RS measurement is used in every year, a change in this gap comes entirely from a change in the administrative record.

Estimation and interpretation. I estimate

$$G_{it,k} = \alpha_{i,k} + \sum_{s \neq 2022-23} \rho_{s,k} \mathbf{1}\{t = s\} + u_{it,k}. \quad (5.10)$$

Each property has its own fixed effect, $\alpha_{i,k}$. The coefficient $\rho_{s,k}$ measures the average within-property change from FY2022–23 to year s . Standard errors are clustered by assessment circle. Small changes indicate stable recording, even when they are statistically different from zero. They do not establish that the record is accurate or identify the effect of the reform. Exact-equality and tolerance-based measures are described in Appendix C.2.

5.5 Measurement and Tax Progressivity

The main distributional comparison holds the tax formula fixed and changes only the physical measurements used to calculate liability. For source $m \in \{\text{admin}, \text{RS}\}$, let T_i^m be the resulting liability. I use the same predicted capital value $Z_i > 0$, property sample and weights w_i in both constructions. Properties retain the same value ranking. Progressivity is therefore measured relative to predicted property value, not household income or independently observed wealth.

Aggregate progressivity. The main measure is the Kakwani index:

$$\mathcal{K}^m = C_T^m - G_Z, \quad \Delta \mathcal{K} = \mathcal{K}^{\text{RS}} - \mathcal{K}^{\text{admin}}. \quad (5.11)$$

Here C_T^m measures how concentrated liabilities are among higher-value properties, and G_Z measures the concentration of property values. A positive index means that liabilities are more concentrated at higher values than property values themselves. A positive $\Delta \mathcal{K}$ means that the RS-input reconstruction is more progressive. Appendix B.3 provides the formula linking this comparison to the model.

Upper-tail effective rates. I also compare the effective rate for the highest-value 10% with the overall rate:

$$R^m(x) = \frac{ATR_{\text{Top } x}^m}{ATR_{\text{All}}^m} - 1, \quad ATR_{\text{Top } x}^m = \frac{\sum_{i \in \text{Top } x} w_i T_i^m}{\sum_{i \in \text{Top } x} w_i Z_i}. \quad (5.12)$$

A negative value means that the high-value group faces a lower effective rate than the population represented by the weights. The reported contrast is

$$\Delta R(10) = R^{\text{RS}}(10) - R^{\text{admin}}(10). \quad (5.13)$$

This compares two normalized rates, each using its own overall rate. It is not a %-point change in the statutory tax rate.

Estimation and interpretation. I calculate both measures from the two liability vectors and use 2,000 assessment-circle bootstrap draws for uncertainty. The full statutory schedule is applied to each source, including when measurement changes exemption status. The comparison does not require every property to remain within one tax band. It estimates the distributional implications of replacing inputs, not the causal effect of reform or the revenue from correcting records. Model-based transport weights also require assumptions about properties outside the observed RS sample; weighting does not remove the coverage limitation described in Section 4.2.

Section 6.6 reports a separate comparison of observed administrative bills and RS-reconstructed liabilities. Observed bills can also reflect transition rules and other billing adjustments, so that comparison cannot isolate physical measurement alone.

6 Results

The central findings concern the continuity of administrative inputs and their distributional consequences under the new formula. Recorded land areas change little across years, while the common-formula comparison produces more progressive liabilities under independently measured inputs. I first establish the measurement basis and its value for targeting verification, then examine recording incentives, persistence and progressivity. The final subsection describes how observed bills and reconstructed liabilities differ across property values. Each exercise reports its estimation sample, reflecting the combinations of administrative, RS and field observations it requires.

6.1 Relative Measurement Performance

Table 1 compares administrative and remote-sensing measurement errors against the same field protocol. For land area, the table contains 2,768 validation properties in 60 assessment circles. Mean-squared error is 742.6 for the administrative record and 320.1 for remote sensing. Equation (5.2) therefore gives $\hat{\lambda}_{\text{land}} = 0.569$, with a 95% assessment-circle bootstrap interval of [0.345, 0.749]. For the covered area, the table contains 2,504 validation properties in 60 assessment circles. Mean-squared error is 19,274,721.7 for the administrative record and 12,163,187.4 for remote sensing. Equation (5.2) therefore gives $\hat{\lambda}_{\text{covered}} = 0.585$, with a 95% assessment-circle bootstrap interval of [0.275, 0.948].

Diagnostic Table 1: Relative measurement performance of remote sensing and the administrative cadastre, against field measurement

	Attribute	
	Land area	Covered area
Properties in the validation sample	2,768	2,504
Tax circles	60	60
Mean squared error, administrative record	742.6	19,274,721.7
Mean squared error, remote sensing	320.1	12,163,187.4
$\delta_{RS} = \text{MSE}(\text{RS}) - \text{MSE}(\text{admin})$	-422.4*** (157.2)	-7,111,534.3*** (2,162,426.2)
Median absolute error, administrative record	1.88	641.88
Median absolute error, remote sensing	0.41	1,199.38
$\lambda = 1 - \text{MSE}(\text{RS})/\text{MSE}(\text{admin})$	0.569	0.585
95% CI (circle cluster bootstrap)	[0.345, 0.749]	[0.275, 0.948]
λ in logs (robustness)	0.306	0.978

The benchmark is the field measurement collected by trained enumerators on site. Accuracy is stated relative to that protocol; neither the site measurement nor remote sensing is assumed to equal latent physical truth without error. δ_{RS} is the coefficient on a remote-sensing indicator in a paired squared-loss regression with property fixed effects. The property fixed effect makes δ_{RS} the mean within-property difference in squared loss, which is identically $\text{MSE}(\text{RS}) - \text{MSE}(\text{admin})$. The omitted source is the administrative record. A negative δ_{RS} means remote sensing is the more accurate source. Standard errors in parentheses, clustered by tax circle. λ is a ratio of means and has no closed-form standard error; its interval is a percentile bootstrap resampling tax circles with replacement, 2,000 replications, seed 20260825. The log specification reweights away from the largest properties and is undefined at zero, so it drops 1 land and 10 covered-area properties. Land area is in marla and covered area in square feet, so the squared errors are not comparable in level across the two columns; λ is scale-free and is comparable.

For land, the estimate implies approximately 57% lower mean-squared error under remote sensing than under administrative measurement, evaluated against the same field protocol. This supports the use of independent measurement as a check on recorded physical inputs. Performance is specific to the attribute and error metric: the covered-area table reports lower RS mean-squared error but a higher median absolute error. Neither source is treated as error-free, and average validation performance does not by itself establish the local identifying assumptions required by the threshold comparison.

6.2 Targeting Verification

Independent measurement is useful for addressing persistent records if its discrepancies identify where a field check is likely to find an error. Figure 6.1 evaluates a list consisting of the 10% of properties with the largest administrative–RS land discrepancies. Of these, 77.9% have a field-confirmed error, compared with 39.5% under random selection. The gain is 38.4 percentage points: approximately 38 additional errors found per 100 inspections. The exercise demonstrates more productive targeting of verification; it does not observe the subsequent correction of records.

The top-decile list identifies 19.7% of all field-confirmed land errors and captures 39.6% of total absolute land error measured in logs. For covered area, precision rises from 43.9% under

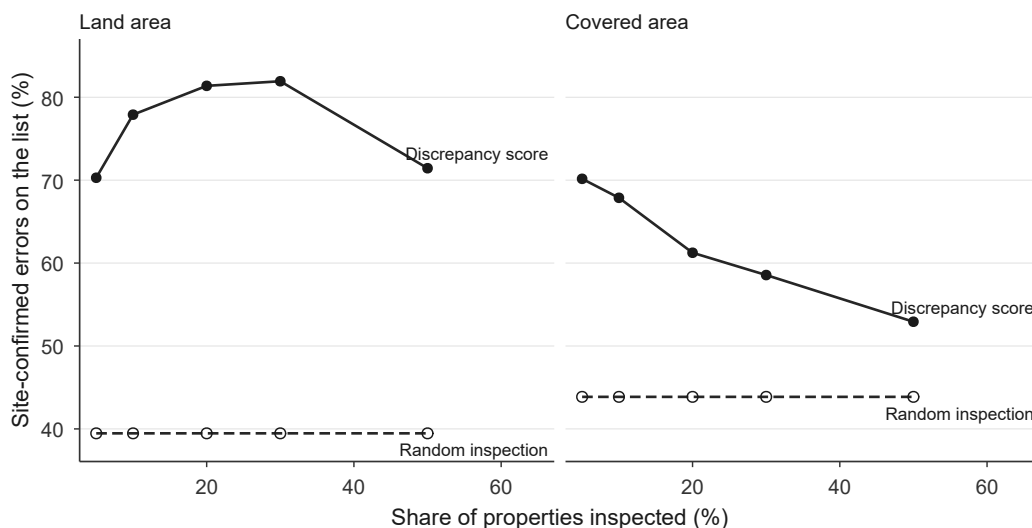


Figure 6.1: **Targeting verification using land-area discrepancies.** Properties are ranked by absolute administrative–RS log differences. Precision is the share with an administrative–field difference exceeding 0.25 absolute log points. Field measurements do not enter the ranking score.

random selection to 67.9% under targeted selection. Full results across inspection shares and error tolerances appear in Tables C.7 and C.8. These comparisons give independent measurement a defined role in the reform process: it can direct attention towards inherited inputs that warrant verification.

6.3 Recording and Tax Incentives

6.3.1 The Land-area Exemption Threshold

The historical exemption provides evidence on why recorded inputs may reflect earlier tax incentives. Figure 6.2 compares administrative placement around five marla in non-A localities, where the size exemption applied, and Category-A localities, where it did not. Panel A displays the administrative-area distributions; Panel B compares placement in the $[3, 5]$ -marla region among properties that RS places in bands above five. Among the 1,893 properties in the widest displayed band, $(5, 8]$ marla, 58.67% in non-A localities fall administratively in $[3, 5]$, compared with 12.74% in Category-A localities. The reported difference is 45.94 percentage points, with a 95% locality-bootstrap interval of $[25.96, 61.51]$. This descriptive upper-side contrast, $\hat{\Delta}_5$, shows that placement in the size range eligible for exemption is more common where that placement changes liability.

Benchmark-confirmed administrative placement around five marla

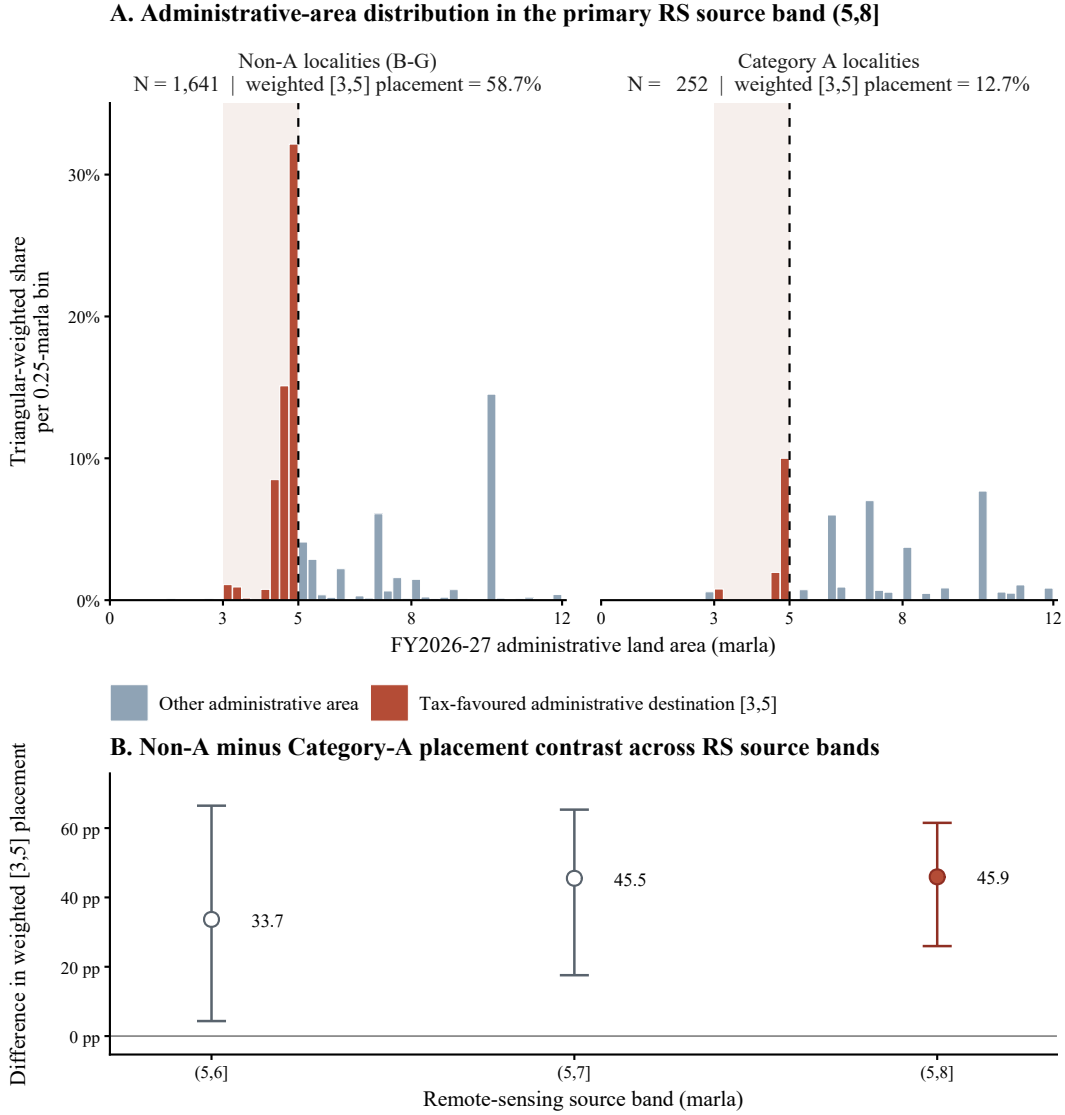


Figure 6.2: **Recorded placement around the land-area exemption threshold.** The sample comprises Lahore residential properties. Panel A shows administrative placement by locality group for RS area in [3, 8] marla. Panel B compares placement in [3, 5] for RS source bands (5, u]. Differences are triangular-distance-weighted non-A minus Category-A contrasts. Capped bars are 95% intervals from 999 locality-cluster bootstrap replications.

Table 6.1 reports the difference in discontinuities defined in Equation (5.7). With a common two-marla bandwidth, the probability of administrative placement at or below five falls by 36.4 percentage points at the benchmark threshold in Category-A localities and rises by 30.3 percentage points in non-A localities. The estimated difference, $\hat{d}_5$, is 66.6 percentage points. The estimate uses 3,153 properties, including 205 in Category-A localities. Under the identifying conditions in Section 5.3, it measures the causal exemption effect on administrative placement at the independent benchmark threshold. It supplies evidence that recording responds to the tax treatment attached to the recorded area, while the subsequent panel exercise asks whether those administrative inputs are updated.

Table 6.1: Benchmark-indexed difference in discontinuities at five marla

	Estimate	Bootstrap s.e.	95% CI	N
<i>Panel A. Difference in discontinuities</i>				
Non-A minus Category-A discontinuity, $\hat{d}_5$	0.666	0.166	[0.342, 0.945]	3,153
Category-A discontinuity	-0.364			205
Non-A discontinuity	0.303			2,948
<i>Panel B. Cross-threshold excess landing at five marla</i>				
Remote sensing, non-A	0.128	0.035	[0.066, 0.201]	—
Category A	0.032	0.031	[0.000, 0.113]	—
Field measurement, non-A	0.076	0.027	[0.033, 0.136]	—
Category A	0.052	0.050	[0.000, 0.184]	—

Notes. Panel A's outcome equals one when FY2026–27 administrative land area is at or below five marla. The running variable is independently measured remote-sensing land area centred at five. The local-linear estimates use a common two-marla bandwidth, a triangular kernel, quadratic bias correction, and predetermined controls for main-road location and log remote-sensing covered area. Assignment cells are circle–locality–block; cells mixing Category-A and non-A properties are excluded. The standard error and percentile interval use 500 joint assessment-circle bootstrap replications. The estimate is benchmark-indexed; a causal exemption interpretation additionally requires a smooth and predetermined benchmark, comparable measurement error and selection across groups, and parallel counterfactual discontinuities.

Panel B reports excess administrative landing at exactly five marla among properties benchmark-located above each comparison point: the landing share at five minus the average landing share at four and six. Remote-sensing and field-measurement benchmarks are reported separately. Intervals use 999 assessment-circle bootstrap replications. These are supporting benchmark-indexed contrasts, not independent causal estimates; the non-A-minus-Category-A triple differences do not exclude zero.

6.3.2 Land Understatement and Potential Tax Liability

The probability curve describes how land understatement varies with potential liability under the earlier rules. Recorded land area is below site-measured area for 40.5% of the 2,606 residential properties in 58 assessment circles. After adjusting for assessment-circle differences, the estimated probability rises from 36.1% at a potential liability of about Rs 3,200 to 43.3% at about Rs 28,700, a difference of 7.2 percentage points. Figure 6.3 shows an increase over the lower part of the liability range followed by a levelling off. Understatement is therefore more common at higher potential liabilities over that lower range. This association describes where discrepancies occur; it does not establish that higher liability causes understatement.

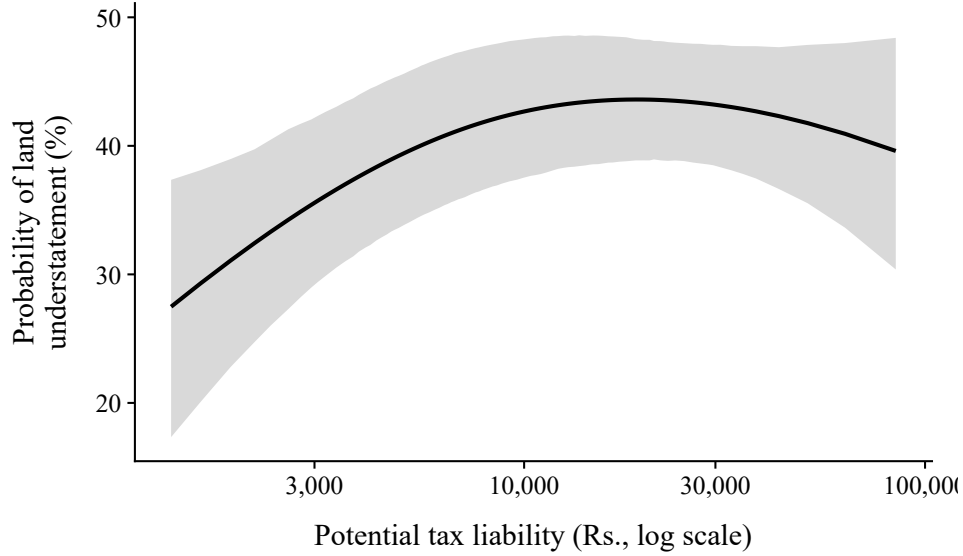


Figure 6.3: **Land understatement and potential tax liability.** The line shows the estimated probability that recorded land area is below site-measured area, from a natural spline in log potential tax liability with assessment-circle fixed effects. Potential liability applies the pre-reform ARV formula to site-measured land and covered areas, using recorded locality and occupancy classifications; it excludes individual allowances, exemptions and rebates and does not measure tax payments. Predictions average over 2,602 properties in 54 circles, holding circle composition fixed. Grey shading gives pointwise 95% intervals from 1,999 circle-bootstrap draws, conditional on the selected spline. The horizontal axis uses a logarithmic scale.

6.4 Persistence of Recorded Attributes

The panel next asks whether a change in valuation rules is accompanied by substantial updating of the recorded inputs. Among 21,816 properties with positive land records in all five years, 92.3% retain exactly the same recorded land area from FY2022–23 through FY2026–27. The recorded quantities therefore show extensive continuity across the period. An unchanged quantity can be correct; this statistic measures stability of the record, not persistence of independently confirmed errors.

Figure 6.4 examines the gap from the fixed RS benchmark. In the balanced land sample of 9,140 properties in 77 assessment circles, the average gap increases by 0.0108 log points over the same period (95% circle-clustered interval: [0.0048, 0.0169]). Its mean changes from -0.060 to -0.049 . The movement is statistically distinguishable from zero but small. Because RS is fixed, it measures a change in recorded area, not repeated observation of physical land area.

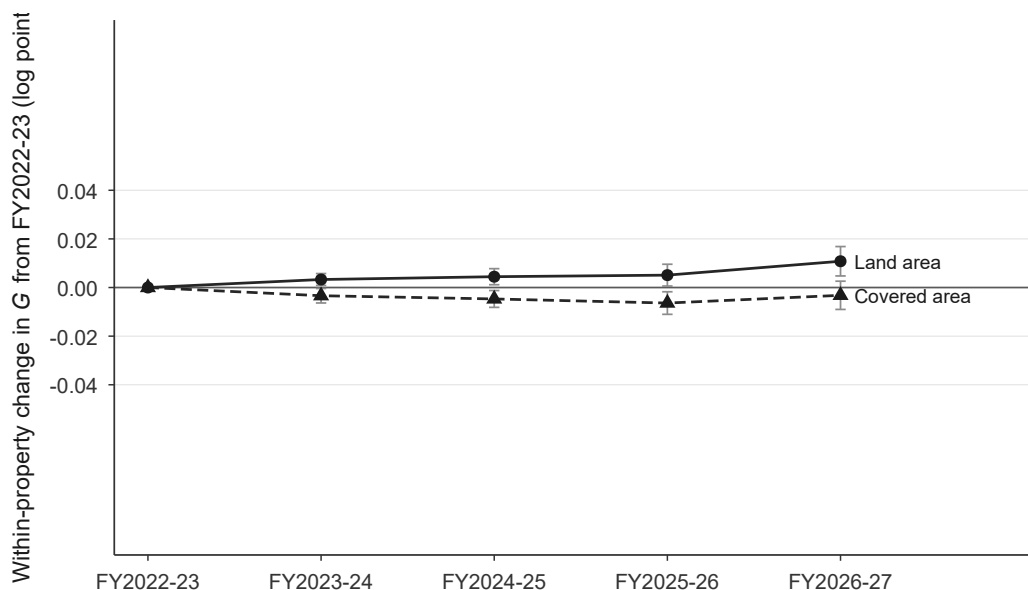


Figure 6.4: **Changes in recorded area relative to a fixed benchmark.** The figure reports within-property changes in $G_{it,k} = \log X_{i,k}^{\text{RS}} - \log X_{it,k}^{\text{admin}}$, relative to FY2022–23. RS measurements are fixed across years. Land is the main persistence outcome; covered area is reported separately. The estimates do not identify a reform effect.

The evidence establishes that the new formula can operate on physical quantities recorded in earlier years. If an inherited quantity is inaccurate, changing the calculation leaves that discrepancy available to affect current liability without any new discretionary adjustment. The panel does not identify why particular records remain unchanged or which discrepancies originate in the earlier exemption. The next exercise establishes why the choice of physical inputs matters for the distribution of liabilities under the new rules.

6.5 Measurement and Tax Progressivity

Holding the valuation rule fixed, do administrative inputs generate the same distribution of liabilities as independently measured inputs? Across 7,956 properties, the Kakwani index is 0.0504 higher under RS inputs. The 95% assessment-circle bootstrap interval is $[0.0141, 0.0992]$, based on 2,000 draws from 10 clusters. The schedule, property-value ranking, denominator and weights are common to both constructions. The difference therefore reflects a change in the concentration of liabilities across the same predicted property-value distribution. It is an index difference, rather than a percentage increase in tax rates or revenue.

Table 6.2: Progressivity under a common tax formula

RS minus administrative inputs	Estimate	95% interval
Kakwani difference, $\Delta\mathcal{K}$	0.0504	[0.0141, 0.0992]
Normalised top-decile difference, $\Delta R(10)$	0.1438	[-0.0468, 0.2381]

The comparison uses 7,956 properties and 2,000 assessment-circle bootstrap draws from 10 clusters. The statutory schedule, sample, weights, property-value denominator and ranking are held fixed across input sources. Intervals are percentile intervals. For each construction, the normalised top-decile outcome is the top-decile effective rate divided by the overall effective rate, minus one. The second row reports the RS-minus-administrative difference in this outcome, not a change in statutory tax rates. Values reproduce the reported common-formula exercise.

The corresponding normalised top-decile difference is 0.1438, with a 95% interval of $[-0.0468, 0.2381]$. Its positive point estimate indicates a higher top-decile rate relative to the overall rate under RS inputs, but the interval includes zero. The aggregate progressivity comparison therefore supplies the clearer distributional finding. The uncertainty estimates remain conditional on inference from 10 assessment-circle clusters.

The result gives the inherited-record argument a distributional consequence: under the same new rules, administrative inputs yield less progressive liabilities than RS inputs relative to the common value benchmark. Proposition 2 explains why that direction depends on the pattern of input discrepancies across property values. Greater progressivity does not itself validate a measurement source. The exercise concerns calculated liabilities under a fixed formula; it neither measures the reform's total causal effect nor the gains realised from an implemented correction programme.

Section 6.6 complements this comparison by showing where observed bills and reconstructed liabilities differ across the property-value distribution.

6.6 Observed Bills and Reconstructed Liabilities

I next examine how observed administrative bills and RS-reconstructed liabilities differ across property values. The concentration curves, upper-tail rates and effective-rate profiles show where the two liability constructions diverge. Unlike the common-formula exercise in Section 6.5, this comparison also reflects differences in billing rules and adjustments. It describes the distribution of observed billing alongside a reconstruction; it does not isolate the contribution of physical measurements.

Upper-tail effective rates. In Figure 6.5, the highest-value decile's effective rate is 14.4% below the overall rate under observed administrative bills and 6.0% below the overall rate under the RS reconstruction. The difference between these normalized profiles is

$$-0.060 - (-0.144) = 0.084. \quad (6.14)$$

The top-decile shortfall relative to the overall rate is therefore narrower under the reconstruction by 8.4 percentage points on this normalised scale. Each construction uses its own overall effective rate, so this is a comparison of relative burdens within each profile, rather than an 8.4-percentage-point increase in the tax rate.

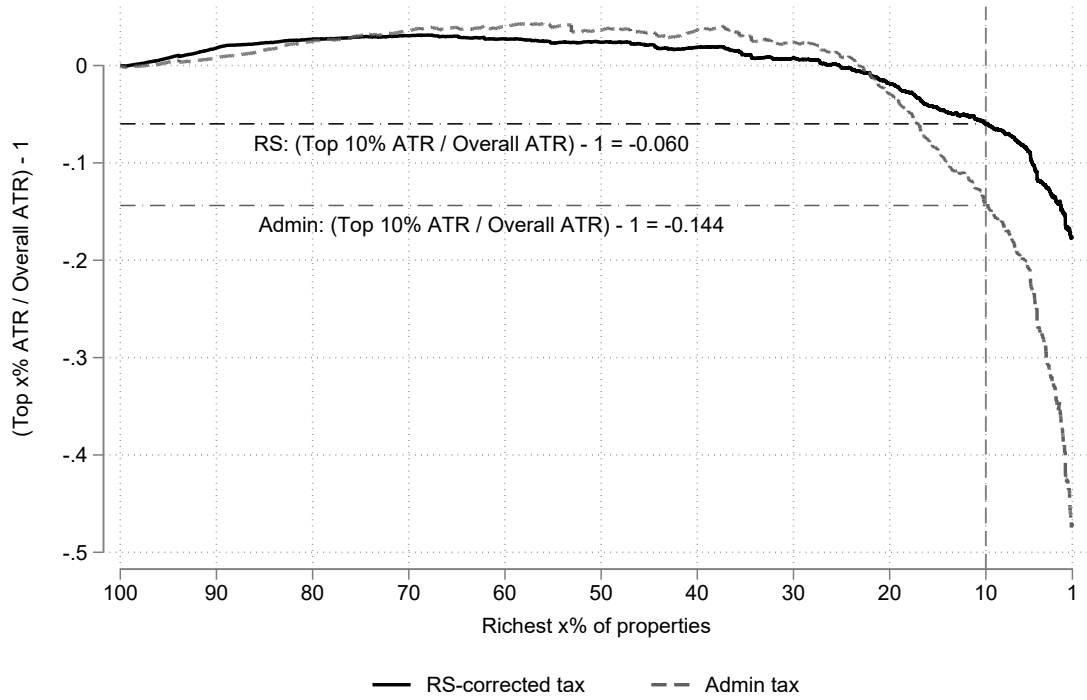


Figure 6.5: **Observed bills and reconstructed liabilities at the top of the value distribution.** The figure compares the top-group effective rate with the overall rate under each liability construction. Properties have the same predicted capital-value ranking. Model-based transport weights target the Lahore residential cadastre. The administrative series uses observed bills; the RS series uses reconstructed liabilities. This comparison does not isolate measured inputs.

Aggregate comparison. The reported Kakwani indices are 0.012 for observed administrative bills and 0.016 for RS-reconstructed liabilities, giving

$$0.016 - 0.012 = 0.004. \quad (6.15)$$

Both indices place liabilities slightly more heavily towards higher-value properties than the distribution of predicted property value itself. The difference of 0.004 describes the two liability constructions shown in the figure. No confidence interval is reported for this comparison. It is a different quantity from the common-formula difference of 0.0504 in Table 6.2, because observed bills can embody different billing rules and adjustments.

Effective-rate profiles. For each construction, the property-level effective rate is

$$e_i^m = T_i^m / Z_i. \quad (6.16)$$

Figure 6.7 shows how the difference varies across the value distribution. Some properties with positive administrative bills are exempt in the reconstruction; at higher values, the administrative profile lies below the reconstructed profile over much of the distribution. These profiles locate differences in liabilities relative to the same predicted value. They do not establish that a household was overcharged or undercharged under the applicable billing rules.

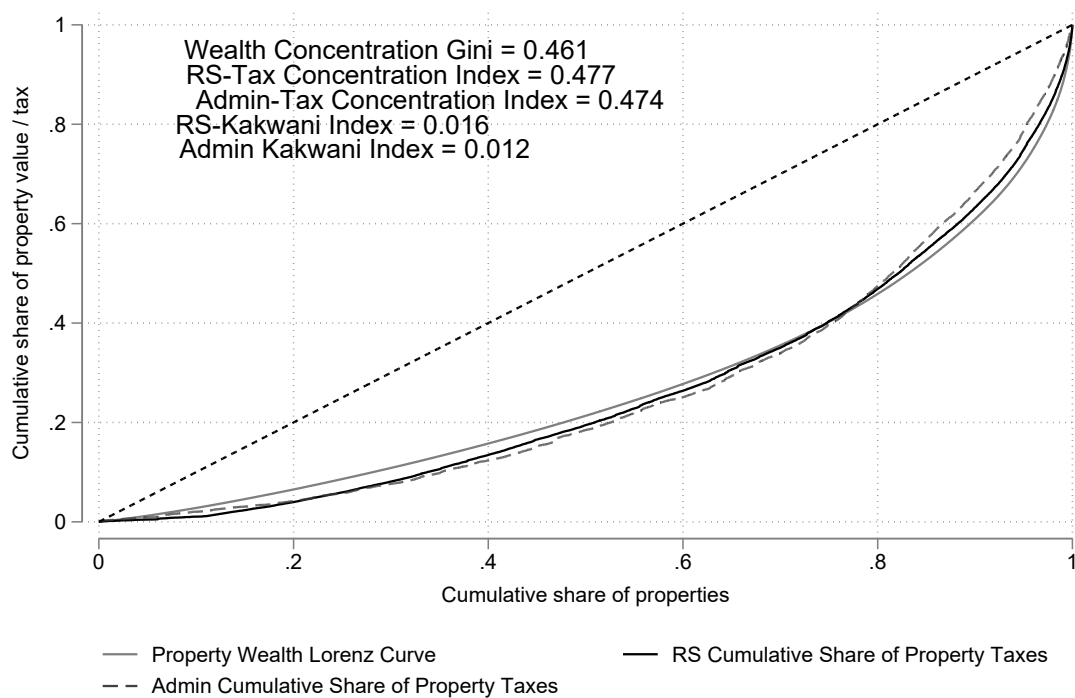


Figure 6.6: **Concentration of observed and reconstructed liabilities.** The figure compares the Lorenz curve of predicted property value with liability concentration curves. The reported Kakwani indices are 0.012 and 0.016. Observed administrative bills and RS-reconstructed liabilities need not follow identical billing rules.

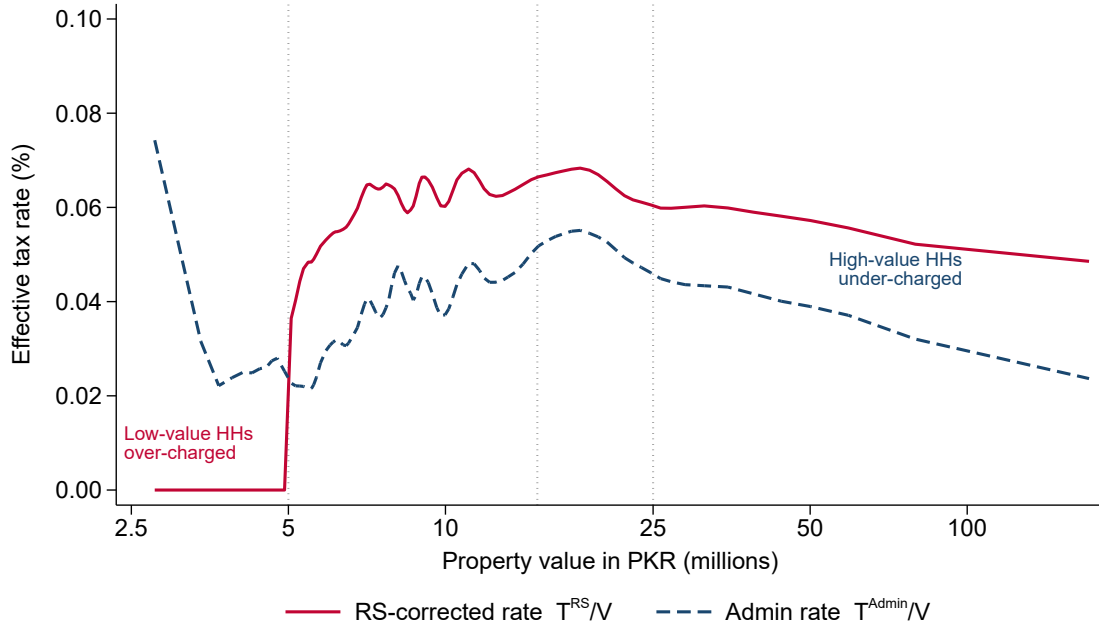


Figure 6.7: **Effective rates under observed bills and RS-reconstructed liabilities.** Both rates use the same predicted capital-value denominator. The profiles describe differences between liability constructions, not a causal effect of reform or a comparison that isolates measurement.

Together, these descriptive profiles show why the distinction between observed bills and reconstructed liabilities matters for interpreting distributional burdens. Section 6.5 supplies the comparison that holds the formula fixed and attributes the resulting distributional difference to the source of physical inputs.

7 Robustness and Scope

7.1 Threshold Comparisons

The difference in discontinuities remains positive along the reported common-bandwidth path in Figure C.8. Group-specific MSE bandwidths give a smaller estimate of 0.433 (bootstrap s.e. 0.171; 95% interval: [0.063, 0.727]). That comparison uses different local supports for the two groups, so the common-bandwidth result remains primary.

Round-number checks are less decisive. Within non-A localities, excess placement at five relative to four and six is positive using either RS or field measurements. However, the corresponding non-A-minus-Category-A triple differences have intervals containing zero. These checks do not provide a separate confirmation of the causal interpretation. Appendix C.6 reports them alongside the heaping diagnostic.

7.2 Measurement, Targeting and Persistence

The land-validation improvement remains positive in logs, with $\hat{\lambda}_L = 0.306$. The targeting gain remains positive over the reported inspection shares from 5 to 50% and error tolerances from 0.10 to 0.50 log points. Tables C.7 and C.8 give the full comparisons. Exact carry-forward complements the log-gap estimates because it records whether the administrative quantity changed

at all. Appendix C.5 reports tolerance-based persistence, zero-valued inputs and the gap distributions.

These checks address measurement definitions and sample restrictions. They do not make field or satellite measurements error-free. Physical change between measurement dates remains especially relevant for covered area.

7.3 Distributional Inference and External Validity

The common-formula comparison is the main distributional result because it isolates the choice of measured inputs. The observed-bill comparison in Section 6.6 is supplementary, not a robustness estimate of the same quantity. Its different point estimates cannot be treated as uncertainty around the common-formula result.

The common-formula bootstrap uses only 10 assessment-circle clusters. Its positive Kakwani interval is informative, but the small cluster count limits precision claims. The RS frame also excludes properties that cannot be reliably delineated from imagery. Model-based weights do not by themselves establish that measurement errors follow the same pattern outside that frame. The distributional results are conditional on the predicted value measure, weighting model and observed coverage.

8 Discussion

The findings identify administrative information as a channel through which earlier arrangements remain relevant after formal tax rules change. Recording responds to exemption incentives under the threshold design's identifying conditions, recorded land quantities show extensive continuity, and alternative physical inputs produce different distributions of liabilities under a common formula. These exercises establish distinct parts of the argument. Together they explain why changing valuation rules can leave the consequences of inherited inputs intact, without estimating how much of the aggregate distributional difference was caused by the earlier exemption.

This form of persistence concerns the information that the administration continues to recognise when determining liability. The panel documents stability of recorded attributes, while the institutional account describes how measurements enter and are amended in the record. Stability alone cannot distinguish accurate records from neglected discrepancies, or identify whether a particular failure to update reflects costs, incentives or approval procedures. The distributional calculation nevertheless establishes why those inputs deserve attention: closer adherence to the new formula does not ensure the same liabilities that would result from alternative physical measurements.

The model clarifies what reform of the calculation can accomplish. Greater verifiability reduces an officer's departure from the administrative benchmark, conditional on the information used by that benchmark. It does not remove inherited input error. Proposition 2 further shows that the effect on relative progressivity depends on how those errors change liabilities across property values. The empirical common-formula result establishes the direction in this setting. The observed-bill profiles add a description of current billing, but they also incorporate differences in rules and adjustments and cannot identify the reform gains lost through inaccurate records.

Technology can help address the information constraint by supplying an independent basis for verification. The targeting exercise shows that large administrative–RS discrepancies

identify more field-confirmed errors than random selection. This is useful precisely because an existing record may otherwise continue to serve as the input to successive calculations. Digitising that same record would preserve its contents; independent measurement supplies a reason to check them. The wider evidence on administrative technology also points to the importance of implementation: electronic information combined with changed financial procedures can improve programme delivery, while digitisation can disrupt other bureaucratic functions (Banerjee et al., 2020; Aman-Rana and Minaudier, 2026).

Realising the value of verification requires a process that converts checked information into an updated record. Such a process should document the measurement and amendment, allow taxpayers to understand and challenge discrepancies, and carry accepted corrections into subsequent billing. The present targeting exercise does not evaluate those administrative responses or their costs. Its contribution is to show where independent information can make verification more productive. The objective is accurate application of the tax rule: correcting an overstatement can reduce a bill, while correcting an understatement can increase it.

9 Conclusion

Formal reform can change how obligations are calculated while retaining the administrative information that determines them. This paper examines that constraint in property taxation. The evidence connects incentive-related recording, continuity of property attributes and the distributional consequences of applying new rules to inherited inputs. Its contribution to institutional persistence is to make this administrative-information channel visible: the consequences of earlier recording can remain relevant even as the valuation system changes.

The distinction has practical implications for reform. A calculation that follows the new rule can still depend on an inaccurate physical record. Improving implementation therefore requires attention to the information the administration accepts and the procedures through which it is renewed. The distributional consequences of correction depend on which properties are mismeasured and in which direction; accurate administration need not raise every bill.

Independent measurement provides a way to identify consequential discrepancies and direct verification. Its value for reform ultimately depends on those checks leading to documented updates that taxpayers can understand and challenge. Changes in formal rules and investment in the capacity to verify their inputs are complementary parts of building an administration that can apply those rules accurately.

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A Figures & Tables

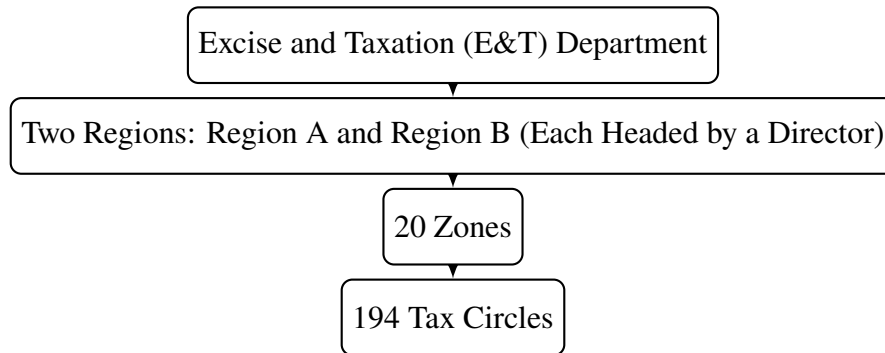


Figure A.1: Flowchart of Property Tax Administration in Lahore

RESIDENTIAL								
Category	Rented				Self			
	Rate of Land Sq. Yd. (In Rs.)		Rate of Covered Area in Sq. Ft. (In Rs.)		Rate of Land Sq. Yd. (In Rs.)		Rate of Covered Area in Sq. Ft. (In Rs.)	
	Upto 500	Exceeding 500	Upto 3000	Exceeding 3000	Upto 500	Exceeding 500	Upto 3000	Exceeding 3000
A	23	18.40	23	18.40	4.60	3.68	4.60	3.68
B	17	13.60	17	13.60	3.40	2.72	3.40	2.72
C	14	11.20	14	11.20	2.80	2.24	2.80	2.24
D	11	8.80	11	8.80	2.20	1.76	2.20	1.76
E	8.20	6.56	8.20	6.56	1.64	1.31	1.64	1.31
F	6.50	5.20	6.50	5.20	1.30	1.04	1.30	1.04
G	4	3.20	4	3.20	0.80	0.64	0.80	0.64

Figure A.2: Pre-reform valuation table used by Excise and Taxation frontline officers to calculate gross annual rental value and property-tax liability. Prescribed rates vary by location category, property use, and occupancy status; for otherwise comparable residential properties, the rate applied to rented property is five times the rate applied to self-occupied property.

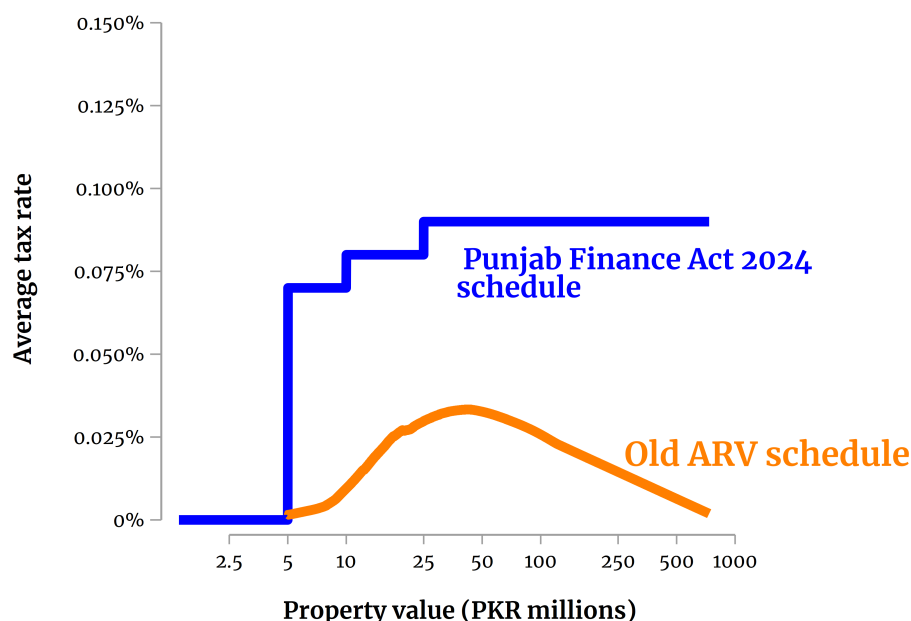


Figure A.3: Average property tax rates under the pre-reform ARV system and the Punjab Finance Act 2024 capital-value schedule. The figure compares the average tax rate implied by Punjab’s pre-reform annual rental value (ARV) system with the capital-value-based schedule introduced by the Punjab Finance Act 2024. The average tax rate is defined as annual property tax liability divided by the property’s capital value. Property values on the horizontal axis are estimated using the official District Collector (DC) valuation rates for 2021–22. The orange line shows the average tax rate under the old ARV system. Under this system, tax liabilities were determined from administratively assessed annual rental values rather than directly from the capital value of the property; the figure expresses these liabilities as a share of DC-rate property values to make the two systems comparable. The blue line shows the statutory schedule introduced by the Punjab Finance Act 2024, under which property tax is instead determined directly from capital value. Properties valued at up to approximately £13,400 (PKR 5 million) are exempt; properties valued above £13,400 and up to approximately £26,800 (PKR 10 million) are taxed at 0.07% of value; properties above £26,800 and up to approximately £67,000 (PKR 25 million) at 0.08%; and properties above £67,000 at 0.09%. Thus, conditional on crossing the exemption threshold, the statutory rates are low and increase only slightly across the three taxable value bands, producing a schedule that is nearly flat in average-rate terms. In contrast, the old ARV system generated a hump-shaped relationship between effective tax rates and capital values, with the tax burden rising among lower- and middle-value properties before declining substantially at higher property values. Sources: Punjab Finance Act 2024 for the capital-value schedule; Punjab Urban Immovable Property Tax (UIPT) MIS, 2021, for pre-reform ARV tax liabilities; Punjab DC valuation tables, 2021–22, for property values. GBP equivalents use an exchange rate of approximately £1 = PKR 373.3.

Property DC Valuation

Property DC Valuation

Land Type

☒ Urban
☐ Rural

Select District

Select Tehsil

Select Town/City

Select Revenue Circle

Select Floor

Select Property Area

Select Land Classification

Select Location

Land Area

DC Rate Rs. 0

DC Sq ft. Rate Rs. 0

Land Value (DC) Rs. 0

FIND LAND RATES **RESET**

(a) Online DC valuation portal

VALUATION TABLE URBAN WAGHA TOWN					
#	AREA	Residential 2018-19		Commercial 2018-19	
		Open Plot Per Marla	Constructed property per sqft	Open Plot Per Marla	Constructed property per sqft
1	ATTOKI AWAN, Bismillah, Al Raheem Garden, Al Ahmed Garden etc (All Residential)	275,000	880	375,000	1,430
2	BAGHBANPURA (ALL TOWN / SOCIETIES)	375,000	880	700,000	1,430
3	BAGRIAN SYEDAN (ALL TOWN / SOCIETIES)	250,000	880	500,000	1,430
4	CHAK RAMPURA (Garison Garden, Rehmat Town etc) (All Residential)	275,000	880	400,000	1,430
5	CHAK DHEERA (ALL TOWN / SOCIETIES)	400,000	880	1,000,000	1,430
6	DARGHAWALA CHOWK TO RING ROAD MEHMOOD BOOTI	500,000	880	750,000	1,430
7	DAVI PURA (ALL TOWN / SOCIETIES)	275,000	880	350,000	1,430
8	FATEH JANG SINGH WALA (ALL TOWN / SOCIETIES)	400,000	880	1,000,000	1,430
9	GOBIND PURA (ALL TOWNS / SOCIETIES)	400,000	880	1,000,000	1,430
10	HANDU, Al Raheem, Masha Allah, Gulshan Dawood Al Ahmed Garden (ALL TOWN / SOCIETIES)	250,000	880	350,000	1,430
11	JALLO, Al Hafeez, IBL Homes, Palm Villas, Aziz Garden etc	250,000	880	500,000	1,430
12	KHEERA, Aziz Garden, Canal Forts, Al Hafeez Garden, Palm Villas (ALL TOWN / SOCIETIES)	250,000	880	500,000	1,430
13	KOT DUNI CHAND Al Karim Garden, Malik Nazir G Garden, Ghous Garden (ALL TOWN / SOCIETIES)	250,000	880	400,000	1,430
14	KOTLI GHASI Hanif Park, Garison Garden, Gulshan e Haider, Moeze Town & New Bilal Gung H Scheme (ALL TOWN / SOCIETIES)	250,000	880	500,000	1,430
15	LAKHODAIR, Al Wadood Garden (ALL TOWN / SOCIETIES)	225,000	880	500,000	1,430
16	LAKHODAIR, Ring Road Par (ALL TOWN / SOCIETIES)	75,000	880	200,000	1,430
17	LIDHAR (ALL TOWN / SOCIETIES)	400,000	880	1,000,000	1,430
18	LEHNA SINGH WALA (ALL TOWNS / SOCIETIES)	400,000	880	1,000,000	1,430
19	MADSUDAN PURA Sharif Garden etc (ALL TOWN / SOCIETIES)	300,000	880	500,000	1,430
20	MAHMOOD BOOTI in side Ring Road & New Shalimar etc (ALL TOWN / SOCIETIES)	350,000	880	500,000	1,430
21	MAHMOOD BOOTI Ring Road Par (ALL TOWN / SOCIETIES)	75,000	880	200,000	1,430
22	MANAWAN, Ghous Garden, Al Hafeez, Ashraf Garden, Atta Town, Tufail Garden & Rizwan Garden (ALL TOWN / SOCIETIES)	250,000	880	700,000	1,430

(b) Underlying DC valuation table

Figure A.4: Publicly accessible District Collector property-valuation system. Panel a shows the publicly accessible online District Collector (DC) valuation portal. A user can select a property's administrative location, land classification, use, floor, and area to retrieve the applicable land rate and implied DC value. Panel b shows an excerpt from the underlying DC valuation list for urban property in Wagha Town, Lahore, which is also publicly available. The list assigns separate benchmark rates to open land, expressed per marla, and constructed property, expressed per square foot, and distinguishes between residential and commercial use. Together, the two panels show that both the interface used to retrieve a property's official valuation and the location-specific rate schedules on which that valuation is based are available to the public. DC values are determined from administratively specified geographic categories and observable property characteristics rather than from a property-specific estimate of market value. Panel b displays the valuation schedule for 2018–19.

(a) Pre-reform assessment, 2023-24

E-mail: etozone1411@punjab.gov.pk
0320-4798990
GOVERNMENT OF THE PUNJAB
FORM P-TA-10 (Current Year)
PT10 Serial #: [REDACTED]
PSID: [REDACTED] (Assessee's Copy)
Circle: [REDACTED]
Rating Area: [REDACTED] Locality: [REDACTED] Year: 2023-24 Issue Date: 20-07-2023 Due Date: 30-09-2023
Property No: [REDACTED] Demand No: [REDACTED]
PIN: [REDACTED]
Owner Name: [REDACTED] Parentage: [REDACTED] Address: [REDACTED]
Land Area (Sq. Yds): 250 Covered Area (Sq. Ft): 2700 Location (Main/Off Road): Main Road Tax 2023-24: 3,894 Remission: 0 Payable 2023-24: 3,894
Category (in Valuation Tables): [REDACTED] Sd/- or Rmtd: [REDACTED]
Tax Payer may avail additional 5% rebate by using ePay application
Total Payable Tax Upto 30-09-2023: 3,699
Amount in Words: Three Thousand Six Hundred Ninety-Nine

What can be verified from the bill

The bill reports land area, covered area, and tax due, but not the valuation rates or a reproducible capital-value base.

(b) Post-reform capital-value assessment, 2026-27

PTS Serial #: [REDACTED] PSID: [REDACTED]
Circle: [REDACTED] (Assessee's Copy)
Rating Area: [REDACTED] Area / Locality: [REDACTED] Year: 2026-27 Issue Date: 07-08-2026 Due Date: 30-09-2026
Property No: [REDACTED] Demand No: [REDACTED]
PIN: [REDACTED]
Owner Name: [REDACTED] Parentage: [REDACTED] Location: [REDACTED]
Land Area (Sq. Yds): 250 Covered Area (Sq. Fts): 2700 DC Base Land Value (Rs.): 10,027,467 DC Base Covered Area Value (Rs.): 5,670,000 Total Property Value: 15,697,467 Total Arrears: 0 Payable: 7,520
Residential Rate: 1,002,750 Commercial Rate: 2,573,250 Industrial Rate: 0 Agriculture Rate: 0
Land Value per Marla (Rs.): 1,002,750 Structure Value per Sq. Fts (Rs.): 2,100
Paid Amount 2023-27: [REDACTED] No Payment: [REDACTED]
Current Tax Upto 30-09-2026: 7,144
Arrears with Surcharge: 0
Payable By 30-09-2026: 7,144
Tax Paid: [REDACTED]
Amount & Bank Receipt No/Date: [REDACTED] Sd/- Name & Seal of Treasury Officer with amount of Tax received: [REDACTED] Treasury Receipt No & Date: [REDACTED] Head of Account: [REDACTED]

What can be verified from the bill

The bill reports the quantities, DC land and structure rates, component values, total property value, and tax due.

Figure A.5: **The capital-value reform made the assessment base reproducible from the tax bill.** Panel (a) shows the pre-reform Assessee's Copy for 2023–24. It reports the property's land area (250 square yards), covered area (2,700 square feet), and final tax demand, but it does not report the valuation rates applied to those quantities or the intermediate assessed property value. A taxpayer or supervisor therefore cannot reconstruct the tax base from the bill itself. Panel (b) shows the post-reform Assessee's Copy for 2026–27. The revised form reports the same physical quantities together with the District Collector (DC) rates—PKR 1,002,750 per marla of residential land and PKR 2,100 per square foot of residential structure—and separately displays the resulting land value (PKR 10,027,467), structure value (PKR 5,670,000), and total property value (PKR 15,697,467). The component values sum to the displayed total, and the published statutory tax schedule can then be applied to check the reported liability. The reform therefore did not merely change the tax base; it made the mapping from recorded property attributes to assessed value observable and externally verifiable. Direct taxpayer, property, account, location, QR-code, and officer identifiers have been redacted.

RS estimation sample and E&T rating jurisdiction, Lahore district

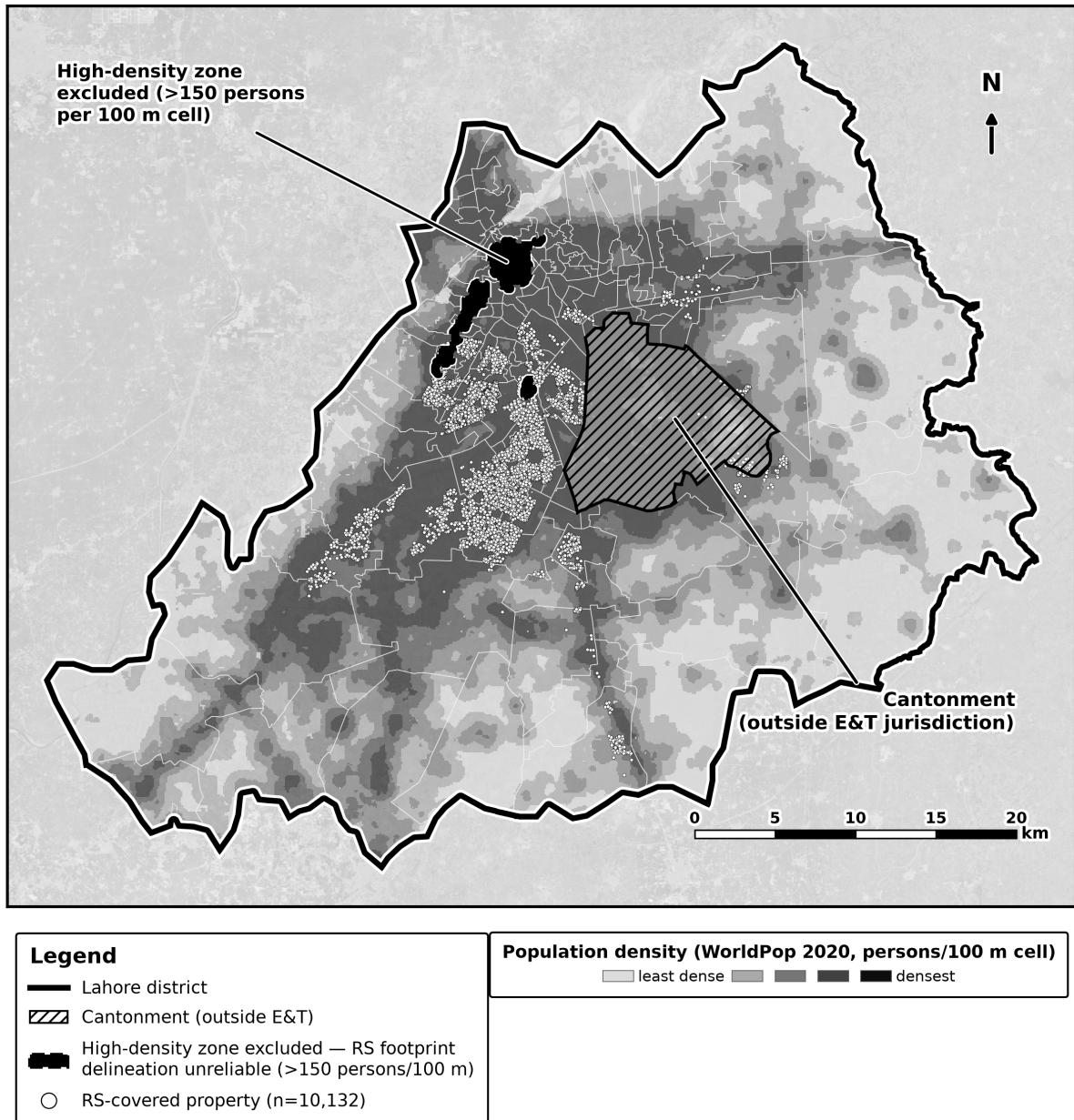


Figure A.6: The remote-sensing estimation sample within Lahore district. Grayscale shading is the WorldPop 2020 population surface (denser cells darker); thin lines are E&T rating jurisdiction. The Lahore Cantonment (hatched) is rated by the Cantonment Board and lies outside E&T jurisdiction; the dashed zone marks the high-density old-city fabric (above 150 persons per 100 m cell) where footprint delineation from imagery is unreliable. Open circles are the $n = 10,132$ RS-covered properties.

B Theoretical Framework: Proofs and Distributional Comparisons

B.1 The Officer's Choice and Proposition 1

The gain over zero departure is

$$\Pi_{ir}(d) - \Pi_{ir}(0) = (\theta - \mu) [T_r(\bar{A}_{ir}) - T_r(\bar{A}_{ir} - d)] - \gamma_r d^2 - \delta p_{ir}(d). \quad (\text{B.1})$$

If $\theta \leq \mu$, this is strictly negative for every $d > 0$, so zero is the unique optimum. Otherwise, the existence assumption in the main text is satisfied, for example, when T_r is lower semicontinuous and p_{ir} is continuous. This permits upward tax jumps with the lower liability assigned at the boundary. Arbitrary discontinuity conventions do not guarantee attainment on a continuous choice set.

Monotonicity. Fix all primitives except γ_r , and suppress property and regime subscripts. Let d_1, d_2 be any global optima at $\gamma_1 < \gamma_2$. Optimality at each parameter value and the identity $\Pi_{\gamma_2}(d) = \Pi_{\gamma_1}(d) - (\gamma_2 - \gamma_1)d^2$ imply

$$(\gamma_2 - \gamma_1)(d_1^2 - d_2^2) \geq 0. \quad (\text{B.2})$$

To obtain this inequality, add $\Pi_{\gamma_1}(d_1) \geq \Pi_{\gamma_1}(d_2)$ and $\Pi_{\gamma_2}(d_2) \geq \Pi_{\gamma_2}(d_1)$. Since departures are nonnegative, $d_2 \leq d_1$. This argument allows multiple optima and discontinuous tax schedules.

Convergence. For $\theta > \mu$, comparison with zero departure yields

$$0 \leq (d_{ir}^*)^2 \leq \frac{(\theta - \mu)[T_r(\bar{A}_{ir}) - T_r(0)]}{\gamma_r}. \quad (\text{B.3})$$

The numerator is finite and fixed. Thus $d_{ir}^* \rightarrow 0$. Substitution into Equation (3.6) proves the assessment limit. At a discontinuity of T , this alone does not establish convergence to $T(V_i + \xi_i^X)$: continuity from below or eventual exact zero departure is needed. Proposition 2 imposes a common affine segment, which supplies the required continuity.

Bounds and tax boundaries. If the entire feasible assessment interval is affine, strict concavity gives the unique global solution

$$d_{ir}^* = \min \left\{ \bar{A}_{ir}, \max \left[0, \frac{(\theta - \mu)\tau}{2H_{ir}} \right] \right\}.$$

With several segments, compare feasible interior candidates, endpoints and actual payoffs at tax boundaries. The full problem need not have a unique optimum. These comparisons hold the benchmark and tax schedule fixed; they do not order departures across an ARV-to-CV reform that changes both.

B.2 Measurement and Proposition 2

With fixed locality rates, the input error has the explicit form

$$\xi_i^X = \rho_{L,l}(a_i^{\text{admin}} - a_i^\dagger) + \rho_{S,l}(s_i^{\text{admin}} - s_i^\dagger). \quad (\text{B.4})$$

This evaluates recorded attributes under a common CV formula. For pre-reform records, it is not an observed ARV assessment. The officer model conditions on these inputs; it does not derive their original recording or subsequent persistence.

Proof of Proposition 2. On each property's stated segment, write $T(x) = c_i + \tau_i x$. Subtracting $T(V_i)$ from $T(V_i + \xi_i^X - d_i^*(\gamma))$ cancels c_i and leaves $\tau_i(\xi_i^X - d_i^*(\gamma))$. Dividing by $V_i > 0$ establishes Equation (3.9). Proposition 1 and continuity then give Equation (3.10). Positive limiting rates allow the ratio to pass to the limit. Dividing each limiting rate by its positive correct-input rate and comparing the two properties gives Equation (3.13).

To verify the sign statement, subtract one:

$$\frac{P_{jk}^\infty}{P_{jk}^{\text{accurate}}} - 1 = \frac{\tau_j \xi_j^X / T(V_j) - \tau_k \xi_k^X / T(V_k)}{1 + \tau_k \xi_k^X / T(V_k)}. \quad (\text{B.5})$$

The denominator is positive because it equals $e_k^\infty / e_k^{\text{accurate}}$. The numerator therefore determines the sign. In proportional segments, $T(V_i) = \tau_i V_i > 0$, so $\tau_i \xi_i^X / T(V_i) = \xi_i^X / V_i$. Equal proportional errors preserve relative progression only under this proportional-segment restriction; affine intercepts can change the comparison.

Tax-boundary crossings and exemptions. When compared measurements cross a boundary, the affine decomposition need not hold. Set current departure to zero exactly and define

$$t_i^I = T(V_i + \xi_i^X), \quad t_i^0 = T(V_i), \quad e_i^I = t_i^I / V_i.$$

For positive liabilities under both constructions, let $f_i = t_i^I / t_i^0$ and $P_{jk}^I = e_j^I / e_k^I$. Without an affine restriction,

$$\frac{P_{jk}^I}{P_{jk}^{\text{accurate}}} = \frac{f_j}{f_k}. \quad (\text{B.6})$$

Under Proposition 2's assumptions, $e_i^I = e_i^\infty$ and $f_i = 1 + \tau_i \xi_i^X / T(V_i)$. Outside those assumptions, evaluate the full schedule rather than substitute the marginal-rate expression. If a reference liability is zero, f_i is undefined. Use liability levels and the aggregate expressions below. Zero liability in a relative-rate denominator also makes that ratio undefined.

B.3 Group Effective Rates and Aggregate Progressivity

Fix a group $\mathcal{B}$, positive population weights w_i , and a common value denominator. Its effective rate is a ratio of totals:

$$E_{\mathcal{B}}^s = \frac{\sum_{i \in \mathcal{B}} w_i t_i^s}{\sum_{i \in \mathcal{B}} w_i V_i}, \quad s \in \{0, I\}. \quad (\text{B.7})$$

When the group's reference liability is positive, $E_{\mathcal{B}}^I / E_{\mathcal{B}}^0$ equals the ratio of its total inherited-input liability to its total correct-input liability. This is not generally an unweighted average of

individual errors. Group comparisons can use the full liability vectors even when individual exemption status changes.

For the Kakwani comparison, normalize $\sum_i w_i = 1$. Let r_i be fixed fractional midranks of the common value distribution and G its unchanged Gini coefficient. For positive total liabilities,

$$C^s = 2 \frac{\sum_i w_i r_i t_i^s}{\sum_i w_i t_i^s} - 1, \quad \mathcal{K}^s = C^s - G.$$

The common Gini cancels, giving the general identity

$$\mathcal{K}^I - \mathcal{K}^0 = 2 \left[\frac{\sum_i w_i r_i t_i^I}{\sum_i w_i t_i^I} - \frac{\sum_i w_i r_i t_i^0}{\sum_i w_i t_i^0} \right]. \quad (\text{B.8})$$

This allows properties to enter or leave taxation.

If every reference-exempt property remains exempt, define $\pi_i^0 = w_i t_i^0 / \sum_h w_h t_h^0$ and $f_i = t_i^I / t_i^0$ on the reference-taxable support. Zero inherited liabilities are allowed. With expectations and covariances taken under π_i^0 ,

$$\mathcal{K}^I - \mathcal{K}^0 = 2 \left[\frac{\text{E}_0[r_i f_i]}{\text{E}_0[f_i]} - \text{E}_0[r_i] \right] = \frac{2 \text{Cov}_0(r_i, f_i)}{\text{E}_0[f_i]}. \quad (\text{B.9})$$

Thus a negative covariance lowers aggregate progressivity, and a positive covariance raises it. A factor f_i decreasing with value rank implies a nonpositive covariance; an increasing factor implies a nonnegative covariance. Individual pairs need not determine the aggregate sign. When reference-exempt properties become taxable, use Equation (B.8).

Empirical interpretation. The comparison must hold the sample, weights, value denominator, ranks and liability schedule fixed. Replacing $V_i^\dagger$ with an empirical reference changes the interpretation to progressivity relative to that reference. Remote-sensing formula values and estimated market values are distinct objects; the empirical section must state which it uses. Field validation supports assessment of measurement quality, but a higher Kakwani index does not itself establish more accurate inputs. Neither identity identifies the causal effect of reform, a change in d_i^* , or the effect of actually correcting records.

C Empirical Construction and Supporting Results

C.1 Measurement Sources and Relative Accuracy

Properties are linked by PIN. Comparisons use a common set of properties observed in the sources required for that exercise. Area units are harmonized before valuation. The common-formula comparison holds use, locality and rates fixed, so it changes measured physical inputs rather than legal classifications.

For source $m \in \{\text{admin}, \text{RS}\}$, the sample MSE and relative improvement are

$$\widehat{\text{MSE}}_k^m = \frac{1}{n} \sum_i (X_{i,k}^m - X_{i,k}^{\text{site}})^2, \quad \widehat{\lambda}_k = 1 - \frac{\widehat{\text{MSE}}_k^{\text{RS}}}{\widehat{\text{MSE}}_k^{\text{admin}}}. \quad (\text{C.10})$$

Both MSEs use the same properties. Let measurement error relative to the unobserved physical

attribute be $\epsilon^m = X^m - X^{true}$. Then

$$E[(X^m - X^{site})^2] = E[(\epsilon^m)^2] + E[(\epsilon^{site})^2] - 2E[\epsilon^m \epsilon^{site}].$$

If the final cross-moment is zero for each source, field error adds the same nonnegative quantity to both MSEs. This preserves their ordering and attenuates the relative MSE difference towards zero. Uncorrelated errors alone are insufficient if their means are nonzero. The assumption concerns the population comparison, not a guarantee about confidence-interval coverage.

Table C.1: Relative measurement performance against field measurements

	Land	Covered area
Properties	2,768	2,504
Assessment circles	60	60
Relative MSE improvement, $\hat{\lambda}_k$	0.569	0.585
95% circle-bootstrap interval	[0.345, 0.749]	[0.275, 0.9485]

Each source is compared with field measurements on the same attribute-specific sample. Positive values indicate lower RS mean-squared error relative to the field protocol. Intervals use 1,000 assessment-circle bootstrap draws. These comparisons do not establish that field measurements are error-free.

C.2 Measurement Dates and Persistence Definitions

RS is a single cross-section from May 2026. Comparing it with administrative records from other dates can combine measurement differences with physical change. Land and covered area are therefore reported separately. Stability in recorded area does not prove that the physical property was unchanged.

Exact persistence requires the same positive recorded quantity in every included year. Tolerance-based persistence allows the changes specified in the accompanying tables. Zero-valued inputs are reported separately because they cannot enter log comparisons. For the gap $G_{it,k}$, RS is fixed, so $G_{it,k} - G_{is,k} = \log X_{is,k}^{admin} - \log X_{it,k}^{admin}$. A stable gap is evidence of stable recording; it does not independently validate either measurement source.

C.3 Formula-value and Liability Accounting

For recorded and comparison attributes evaluated at the same locality rates, the formula-value difference is

$$V_i^{admin} - V_i^m = \rho_{L,l}(a_i^{admin} - a_i^m) + \rho_{S,l}(s_i^{admin} - s_i^m), \quad m \in \{RS, site\}.$$

This is the empirical counterpart of Equation (B.4); a measured comparison source does not become physical truth by construction. The common-formula exercise applies the same full statutory schedule to each source. It excludes transition caps and payments, and allows source differences to change exemption status. Observed administrative bills are used only in the separate descriptive comparison below.

C.4 Audit-score Evaluation

For the attribute-specific score and field-confirmed error indicator in Equation (5.5), let $S_{i,k}(q)$ indicate membership in the highest-scoring share q . Precision, random-selection precision and their difference are

$$\begin{aligned}\hat{\pi}_k(q, t) &= \frac{\sum_i S_{i,k}(q) E_{i,k}(t)}{\sum_i S_{i,k}(q)}, \quad \hat{E}_k(t) = \frac{1}{n} \sum_i E_{i,k}(t), \\ \hat{\Delta}_k(q, t) &= \hat{\pi}_k(q, t) - \hat{E}_k(t).\end{aligned}\tag{C.11}$$

The null of no targeting gain is $\Delta_k = 0$, not zero precision. Recall is the fraction of all field-confirmed errors found by the list. Absolute-error capture is the fraction of total $|\log X^{admin} - \log X^{site}|$ on that list.

The reported grid uses $q \in \{0.05, 0.10, 0.20, 0.30, 0.50\}$ and $t \in \{0.10, 0.25, 0.50\}$. The score is fixed without using field outcomes; this exercise does not estimate a prediction model on the validation outcomes. Each of 1,000 circle-bootstrap draws recalculates the ranking and gain. The reported two-sided bootstrap tail share is twice the smaller fraction of draws on either side of zero; it is not an exact randomization test. Because the score and outcome share administrative measurements, correlated errors across sources can affect their relationship. Interpretation remains relative to the field protocol.

For covered area, the top-decile list finds 15.4% of all field-confirmed errors and captures 17.6% of absolute log error. These summarize targeting performance rather than subsequent administrative action.

C.5 Detailed Persistence and Targeting Results

Tables C.5 and C.6 report gap distributions and within-property changes. Tables C.7 and C.8 report precision, recall and absolute-error capture across inspection shares and tolerances.

C.6 Descriptive Placement and Threshold Checks

For properties above the RS threshold, define

$$B_i^5 = \mathbf{1}\{3 \leq A_i \leq 5\}, \quad v_i(h) = \left(1 - \frac{X_i - 5}{h}\right) \mathbf{1}\{5 < X_i \leq 5 + h\}.\tag{C.12}$$

The main descriptive band uses $h = 3$. Its contrast is

$$\Delta_5 = \frac{\sum_i v_i(3) N_i B_i^5}{\sum_i v_i(3) N_i} - \frac{\sum_i v_i(3) (1 - N_i) B_i^5}{\sum_i v_i(3) (1 - N_i)}.\tag{C.13}$$

This compares weighted favourable-placement shares among properties that RS places in $(5, 8]$. The current sample contains 1,893 properties and gives a contrast of 45.94% points. The bootstrap resamples localities 999 times. It is a different comparison from the two-sided difference in discontinuities and should not be interpreted as the same estimand.

The administrative-to-benchmark pile-up ratio at five is 5.52 in non-A localities and 1.59 in Category A. Their estimated difference is 3.94 (95% interval: $[1.60, 7.28]$). Seven marla also shows a positive group difference and four a negative one. Thus heaping at common plot sizes cannot, on its own, isolate the exemption incentive.

Table C.2: Cadastral persistence: percentage of properties whose recorded input changes between adjacent administrative vintages

Cadastral input	FY23–24	FY24–25	FY25–26	FY26–27
Land area (marla)	2.29	0.63	0.70	4.87
Covered area (sq. ft)	4.07	1.21	1.29	2.06
Storeys	1.61	0.73	1.00	6.09
Location class	0.61	0.26	0.02	0.45
Residential locality category	0.11	0.00	0.02	0.60
Properties in both vintages	22,054	22,054	22,054	22,054

Each column reports the change from the preceding administrative vintage into the fiscal year named in the header.

Each cell is the percentage of properties observed in both vintages whose recorded value is not exactly equal across them. Exact equality is the primary persistence definition; Table C.3 reports tolerance-based definitions.

Location class and locality category are compared after normalising case and punctuation, because the cadastre spells the location class “Main Road” in some vintages and “Main-Road” in others. Comparing the raw strings would record a change where none occurred. The literal entry “NULL” is treated as missing, not as a category.

Across all five inputs and every adjacent pair, 82.8 per cent of the 22,054 properties observed in at least one pair show no change in any input. Restricting to the two inputs the capital-value formula prices, land area and covered area, the figure is 88.0 per cent of 22,054 properties.

Sample: properties recorded in Lahore district. Persistence is not accuracy: an input carried forward unchanged may be carried forward wrong.

Table C.3: Cadastral persistence under alternative tolerances: percentage of properties whose recorded input changes

Cadastral input	Definition of “changed”	FY23–24	FY24–25	FY25–26	FY26–27
Land area (marla)	Exact	2.29	0.63	0.70	4.87
	$ \Delta \log X > 0.01$	1.81	0.59	0.64	3.70
	$ \Delta \log X > 0.05$	1.59	0.51	0.61	3.06
	$ \Delta \log X > 0.10$	1.44	0.46	0.58	2.78
Covered area (sq. ft)	Exact	4.07	1.21	1.29	2.06
	$ \Delta \log X > 0.01$	3.60	1.06	1.06	1.89
	$ \Delta \log X > 0.05$	3.40	0.98	1.03	1.83
	$ \Delta \log X > 0.10$	3.09	0.94	0.96	1.76
Storeys	Exact	1.61	0.73	1.00	6.09
	$ \Delta \log X > 0.01$	1.33	0.46	0.61	6.05
	$ \Delta \log X > 0.05$	1.33	0.46	0.61	6.05
	$ \Delta \log X > 0.10$	1.33	0.46	0.61	6.05

Each column reports the change from the preceding administrative vintage into the fiscal year named in the header.

The same tolerance grid is applied to every fiscal-year transition. The log-change definitions are undefined when either vintage records a zero, so those properties are excluded from the tolerance rows and counted in Table C.4 rather than being recoded as unchanged.

Exact equality is the primary definition reported in Table C.2.

Table C.4: Zero-valued cadastral inputs, by transition

Cadastral input	Transition	Observed in both	Zero in earlier vintage	Zero in later vintage
Land area (marla)	FY23–24	22,054	159	167
Land area (marla)	FY24–25	22,054	167	168
Land area (marla)	FY25–26	22,054	168	178
Land area (marla)	FY26–27	22,005	178	181
Covered area (sq. ft)	FY23–24	22,054	204	226
Covered area (sq. ft)	FY24–25	22,054	226	219
Covered area (sq. ft)	FY25–26	22,054	219	208
Covered area (sq. ft)	FY26–27	22,054	208	203
Storeys	FY23–24	22,054	195	181
Storeys	FY24–25	22,054	181	217
Storeys	FY25–26	22,054	217	168
Storeys	FY26–27	21,826	10	0

Zero recorded land area, covered area or storeys is reported separately rather than silently recoded, because a zero may itself be an administrative measurement failure rather than a true value.

Table C.5: Recorded–benchmark disagreement in land area: $G_{it} = \log X_i^{RS} - \log X_{it}^{admin}$

Fiscal year	N	Mean G	Median G	Interquartile range	Share $G > 0$ (%)	Within-property change	s.e.
FY2022–23	8,387	-0.173	-0.075	-0.32 to 0.10	38.9	— (base)	
FY2023–24	8,385	-0.170	-0.074	-0.32 to 0.10	39.1	0.0033***	(0.0010)
FY2024–25	8,386	-0.168	-0.073	-0.32 to 0.10	39.2	0.0046***	(0.0015)
FY2025–26	8,386	-0.167	-0.072	-0.31 to 0.11	39.4	0.0059***	(0.0020)
FY2026–27	8,383	-0.162	-0.070	-0.31 to 0.11	39.7	0.0108***	(0.0028)

A positive G means the administrative record implies a lower quantity than the independent remote-sensing measurement.

The final two columns report β_s from $G_{it} = \alpha_i + \sum_s \beta_s \mathbf{1}\{t = s\} + \varepsilon_{it}$ with property fixed effects, standard errors clustered by tax circle (76 circles, 8,387 properties).

Because the remote-sensing measurement is a single 2026 cross-section and is held fixed across the panel, β_s measures movement of the ADMINISTRATIVE RECORD against a stationary benchmark. It is not a change in the physical property, and it is not the effect of any policy.

The omitted base year is FY2022–23. The draft names FY2021–22; that vintage is not present in the administrative panel.

Sample: properties with a remote-sensing measurement and a strictly positive administrative record. 41,933 property-year observations have both sources recorded; 41,927 remain after dropping non-positive values, for which the log gap is undefined.

Table C.6: Recorded–benchmark disagreement in covered area: $G_{it} = \log X_i^{RS} - \log X_{it}^{admin}$

Fiscal year	N	Mean G	Median G	Interquartile range	Share $G > 0$ (%)	Within-property change	s.e.
FY2022–23	8,387	0.166	0.189	-0.18 to 0.54	63.4	— (base)	
FY2023–24	8,363	0.161	0.186	-0.18 to 0.54	63.2	-0.0034**	(0.0015)
FY2024–25	8,369	0.159	0.184	-0.19 to 0.54	63.1	-0.0047***	(0.0018)
FY2025–26	8,364	0.158	0.183	-0.19 to 0.53	63.1	-0.0064***	(0.0024)
FY2026–27	8,365	0.160	0.184	-0.19 to 0.54	63.2	-0.0032	(0.0030)

A positive G means the administrative record implies a lower quantity than the independent remote-sensing measurement.

The final two columns report β_s from $G_{it} = \alpha_i + \sum_s \beta_s \mathbf{1}\{t = s\} + \varepsilon_{it}$ with property fixed effects, standard errors clustered by tax circle (76 circles, 8,387 properties).

Because the remote-sensing measurement is a single 2026 cross-section and is held fixed across the panel, β_s measures movement of the ADMINISTRATIVE RECORD against a stationary benchmark. It is not a change in the physical property, and it is not the effect of any policy.

The omitted base year is FY2022–23. The draft names FY2021–22; that vintage is not present in the administrative panel.

Sample: properties with a remote-sensing measurement and a strictly positive administrative record. 41,935 property-year observations have both sources recorded; 41,848 remain after dropping non-positive values, for which the log gap is undefined.

Table C.7: Audit targeting on the administrative–remote-sensing discrepancy in land area: does the score concentrate site-confirmed error?

		Share of properties inspected				
		5%	10%	20%	30%	50%
<i>Error threshold</i> $ \Delta \log X > 0.10$	<i>base rate</i> 66.5%					
Precision (%)		81.2	84.4	86.4	87.7	85.0
Gain over random inspection (pp)		14.7***	18.0***	20.0***	21.2***	18.6***
standard error		(3.9)	(2.2)	(2.2)	(1.7)	(1.6)
Recall (%)		6.1	12.7	26.0	39.6	63.9
Absolute error captured (%)		23.0	39.6	58.3	70.8	84.9
<i>Error threshold</i> $ \Delta \log X > 0.25$	<i>base rate</i> 39.5%					
Precision (%)		70.3	77.9	81.4	81.9	71.4
Gain over random inspection (pp)		30.8***	38.4***	41.9***	42.5***	32.0***
standard error		(7.4)	(4.3)	(3.6)	(2.4)	(1.6)
Recall (%)		8.9	19.7	41.2	62.3	90.5
Absolute error captured (%)		23.0	39.6	58.3	70.8	84.9
<i>Error threshold</i> $ \Delta \log X > 0.50$	<i>base rate</i> 24.8%					
Precision (%)		67.4	76.1	77.9	72.9	46.9
Gain over random inspection (pp)		42.6***	51.3***	53.1***	48.1***	22.1***
standard error		(7.3)	(4.2)	(3.5)	(3.0)	(2.4)
Recall (%)		13.5	30.6	62.7	88.1	94.5
Absolute error captured (%)		23.0	39.6	58.3	70.8	84.9
Properties inspected		138	276	553	830	1,383

Properties are ranked by the discrepancy score $Q_i = |\log X_i^{RS} - \log X_i^{admin}|$ and the highest-scoring share is inspected. The site measurement never enters the score; it appears only in the outcome.

A property is a site-confirmed error when the administrative record differs from the field measurement by more than the stated tolerance in absolute log points. Precision is the share of the inspected list that is a confirmed error. Under random inspection precision equals the base rate, which is reported in each panel heading, so the gain row is the operationally meaningful quantity.

Recall is the share of all confirmed errors that the list finds. Absolute error captured is the share of the sample's total $|\Delta \log X|$ that sits on the inspected list, which weights a large error more heavily than a marginal one.

Standard errors from a circle cluster bootstrap, 1,000 replications, seed 20260825, resampling the 60 tax circles in the validation sample. Stars from the bootstrap distribution of the gain: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The score is a fixed function of the two administrative-side sources with no parameter estimated on the validation data, so no cross-fitting is required. Any tuned score would need circles partitioned before evaluation.

The error is defined on the recorded attribute rather than on a formula value, following the chapter's own rule that outcomes are defined on attributes because the statutory schedule is not recoverable for the pre-reform period.

Table C.8: Audit targeting on the administrative–remote-sensing discrepancy in covered area: does the score concentrate site-confirmed error?

	Share of properties inspected				
	5%	10%	20%	30%	50%
<i>Error threshold $\Delta \log X > 0.10$</i>	<i>base rate 72.7%</i>				
Precision (%)	89.5	87.1	83.1	81.7	78.3
Gain over random inspection (pp)	16.8***	14.4***	10.4***	8.9***	5.5***
standard error	(3.5)	(2.3)	(1.9)	(1.4)	(1.1)
Recall (%)	6.1	12.0	22.8	33.7	53.8
Absolute error captured (%)	9.6	17.6	29.9	40.7	59.9
<i>Error threshold $\Delta \log X > 0.25$</i>	<i>base rate 43.9%</i>				
Precision (%)	70.2	67.9	61.2	58.6	52.9
Gain over random inspection (pp)	26.3***	24.0***	17.4***	14.7***	9.1***
standard error	(4.9)	(3.7)	(2.5)	(2.0)	(1.2)
Recall (%)	8.0	15.4	27.9	40.0	60.3
Absolute error captured (%)	9.6	17.6	29.9	40.7	59.9
<i>Error threshold $\Delta \log X > 0.50$</i>	<i>base rate 19.3%</i>				
Precision (%)	45.2	44.2	36.5	31.1	25.7
Gain over random inspection (pp)	25.8***	24.9***	17.2***	11.8***	6.4***
standard error	(5.2)	(3.8)	(2.9)	(2.0)	(1.1)
Recall (%)	11.6	22.8	37.8	48.3	66.6
Absolute error captured (%)	9.6	17.6	29.9	40.7	59.9
Properties inspected	124	249	498	748	1,247

Properties are ranked by the discrepancy score $Q_i = |\log X_i^{RS} - \log X_i^{admin}|$ and the highest-scoring share is inspected. The site measurement never enters the score; it appears only in the outcome.

A property is a site-confirmed error when the administrative record differs from the field measurement by more than the stated tolerance in absolute log points. Precision is the share of the inspected list that is a confirmed error. Under random inspection precision equals the base rate, which is reported in each panel heading, so the gain row is the operationally meaningful quantity.

Recall is the share of all confirmed errors that the list finds. Absolute error captured is the share of the sample's total $|\Delta \log X|$ that sits on the inspected list, which weights a large error more heavily than a marginal one.

Standard errors from a circle cluster bootstrap, 1,000 replications, seed 20260825, resampling the 57 tax circles in the validation sample. Stars from the bootstrap distribution of the gain: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The score is a fixed function of the two administrative-side sources with no parameter estimated on the validation data, so no cross-fitting is required. Any tuned score would need circles partitioned before evaluation.

The error is defined on the recorded attribute rather than on a formula value, following the chapter's own rule that outcomes are defined on attributes because the statutory schedule is not recoverable for the pre-reform period.

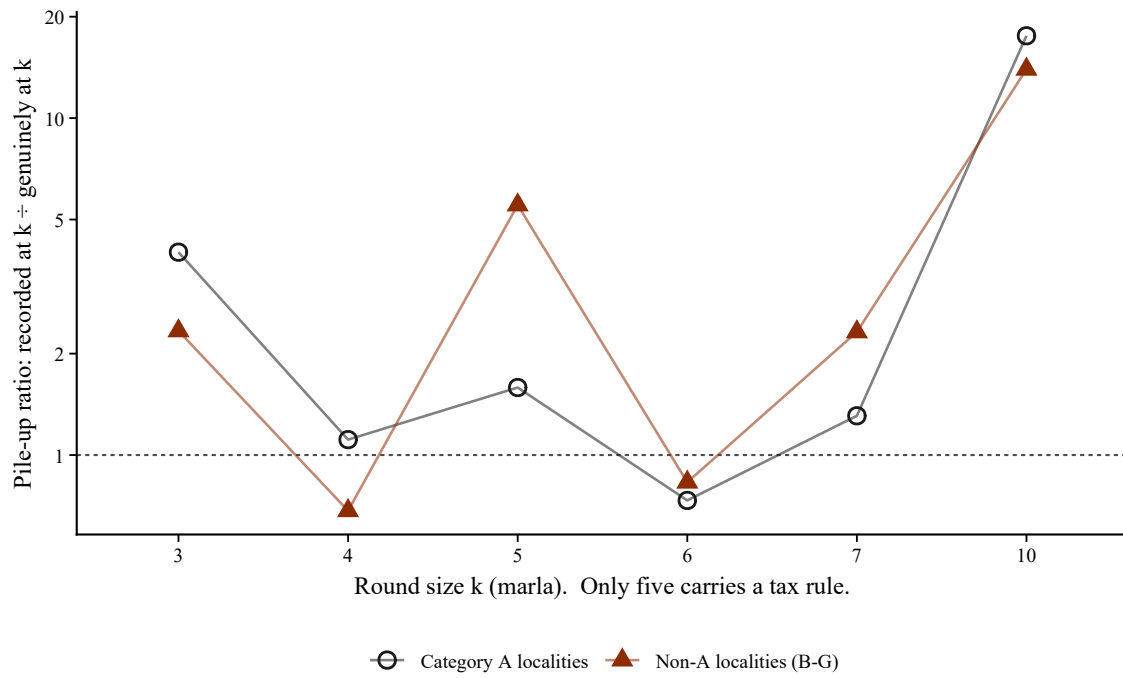


Figure C.7: **Administrative pile-up relative to benchmark mass at common plot sizes.** For each size k , the ratio is the share recorded administratively at k divided by the share that the benchmark places within 0.125 marla of k . A value of one denotes no excess administrative mass. Five marla is both a common plot size and the statutory exemption threshold; differences at the non-statutory sizes show why the figure is treated as a diagnostic rather than a separate causal estimate. The vertical scale is logarithmic.

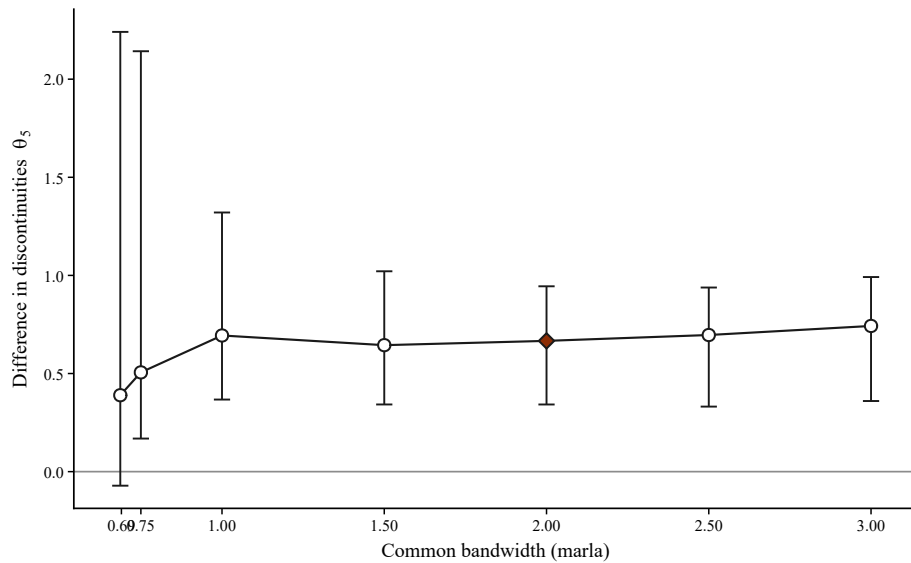


Figure C.8: **Common-bandwidth sensitivity.** The two-marla specification is the main estimate; the remaining points show the bandwidth path.

Within non-A localities, excess placement at five relative to four and six is 0.128 using RS (95% interval: [0.066, 0.201]) and 0.076 using field measurement ([0.033, 0.136]). The non-A-minus-Category-A triple differences are 0.096 ([-0.010, 0.181]) and 0.025 ([-0.112, 0.108]), respectively. Neither triple-difference interval excludes zero. These checks qualify the threshold evidence rather than independently establishing causality.

C.7 Land Understatement: Construction and Estimation

Potential tax liability applies the gross pre-reform ARV formula to site measurements:

$$K_i = 0.05 \times 12 \times r_i [25\ell_i^{site} \omega_L(25\ell_i^{site}) + c_i^{site} \omega_C(c_i^{site})]. \quad (C.14)$$

Here ℓ_i^{site} is land in marla, $25\ell_i^{site}$ is land in square yards, and c_i^{site} is covered area in square feet. The monthly residential rental rates for categories A–G are 23, 17, 14, 11, 8.2, 6.5 and 4 rupees, respectively; the self-occupied rate is one fifth of the rented rate. The rate r_i uses recorded locality and occupancy classifications. The factors 12 and 0.05 annualise rental value and apply the five per cent tax rate. The multiplier ω_L equals 0.8 at or above 500 square yards (20 marla), and one below; ω_C equals 0.8 at or above 3,000 covered square feet, and one below. Each multiplier applies to the whole corresponding component. The calculation excludes individual allowances, exemptions, rebates, arrears and enforcement. Land understatement is measured by $D_i^A = \mathbf{1}\{\ell_i^{site} > \ell_i^{admin}\}$.

Sample and estimation. The sample contains residential properties in Lahore matched to FY2022–23 administrative data, with a valid occupancy indicator, a locality category from A to G, nonblank circle and locality identifiers, positive land measurements from both sources, and nonnegative covered-area measurements from both sources. These restrictions give 2,606 properties in 58 circles. The fixed-effect regression omits singleton circles, leaving 2,602 properties in 54 circles. The Bayesian information criterion selects two spline degrees of freedom from candidates two through six. The figure evaluates the model over the full sample’s 1st to 99th percentiles of potential liability. At each liability, I average predictions over properties in the estimation sample using their observed assessment-circle shares. The tax amounts reported in the text correspond to the full sample’s 10th and 90th percentiles.

For confidence intervals, I resample complete circles 1,999 times, refit the model using the selected spline basis, and average predictions over the resampled properties. The 2.5th and 97.5th percentiles of these predictions give pointwise intervals; they do not incorporate uncertainty from selecting spline flexibility. The spline coefficients are jointly different from zero ($p = 0.006$, using circle-clustered covariance). This test concerns variation along the full curve, rather than the particular difference between the two liabilities reported in the text. Site-measured land enters both potential liability and the understatement indicator, so the association does not by itself identify strategic assessment.

D Limits of Interpretation

Reform effects. The analysis does not provide an untreated comparison group for the capital-value reform. The timing of full conversion depends on the required increase from the earlier liability. Differences by conversion timing can therefore combine baseline differences, transition exposure and calendar-time changes. They do not identify the causal effect of conversion or a change in d_i^* .

Measurement and coverage. Field and RS measurements can both contain error. Differences between their dates and administrative recording dates can also reflect physical change. The RS frame excludes dense and irregular properties that cannot be delineated reliably. Normalized balance differences and transport weights do not prove that unobserved measurement errors behave similarly outside this frame.

Matching and interpretation. PIN linkage establishes a common recorded property identifier, but does not validate its physical boundaries, classification or area. The results concern recording patterns, persistence, field-relative measurement performance, targeting and formula-implied liabilities. They do not identify informal payments, household welfare, realized collections or revenue buoyancy. A higher progressivity index is not itself evidence of greater measurement accuracy.

E Additional figures

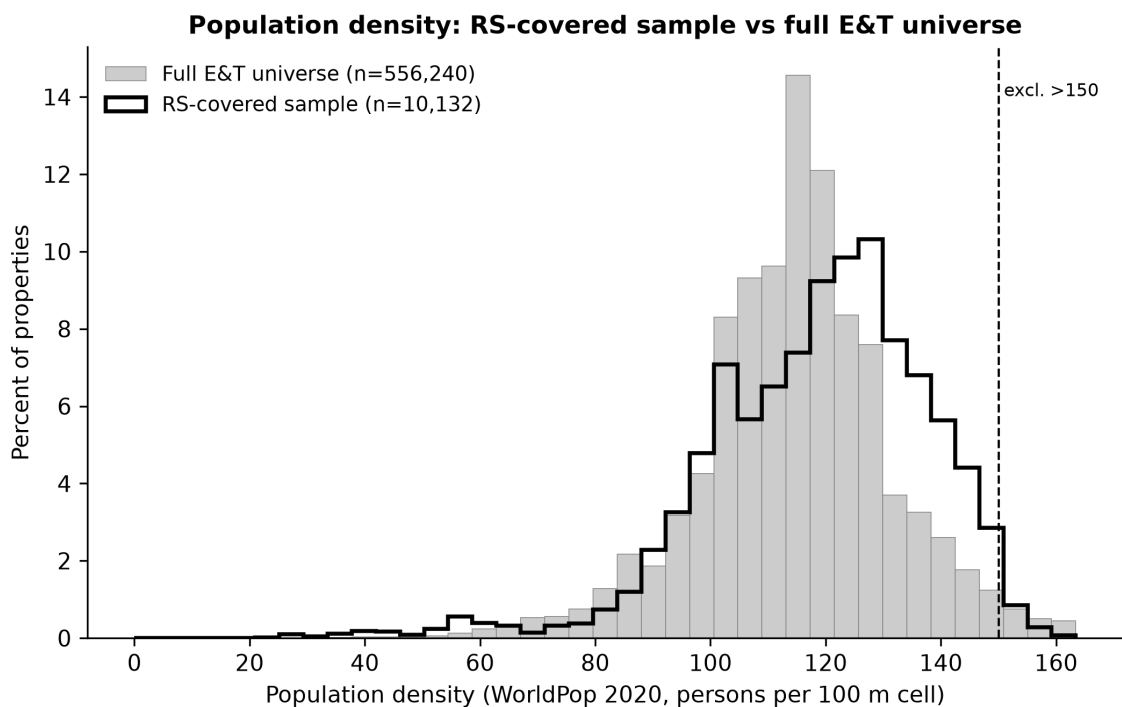


Figure E.9: Population density of the RS-covered sample against the full E&T universe (% of properties per bin). Density is WorldPop 2020 persons per 100 m cell; for register properties without coordinates the value is imputed from the fifteen geocoded properties with the most similar land and covered area, which produces the concentration of universe mass near the distribution's centre. The dashed line marks the 150-person exclusion threshold of Section 4.2.